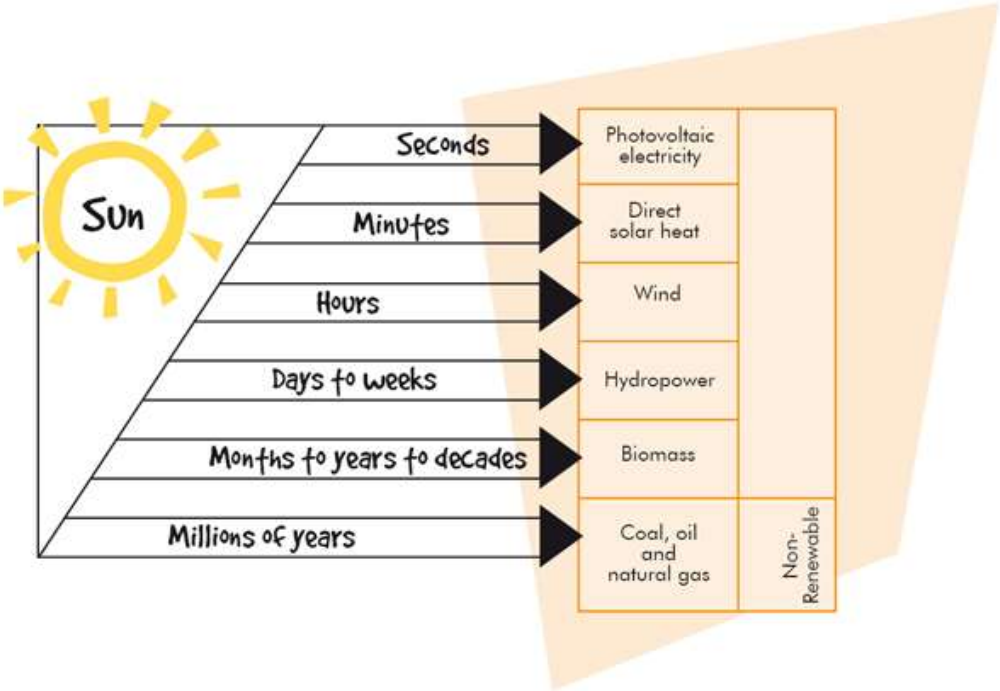
III. Wind Energy

WIND ENERGY

- Wind energy is the kinetic energy of air moving from one place to another in the form of wind.
- Wind is created as the results of uneven heating of the earth by the sun:
  - Warm air rises leaving a vacuum behind
  - Cooler surrounding air moves to fill the vacuum creating wind
- Hence the wind energy is essentially another form of SOLAR energy (2-3% of radiation reaching the earth is converted to wind)



HISTORY OF WIND ENERGY

- Wind energy propelled boats along the Nile River as early as 5000 B.C.
- By 200 B.C., simple windmills in China were pumping water, while they were grinding grain in Persia and the Middle East.
- By the 11th century, people in the Middle East were using windmills extensively for food production; returning merchants and crusaders carried this idea back to Europe.
- The Dutch refined the windmill and adapted it for draining lakes and marshes
- Settlers took this technology to the New World in the late 19th century, they began using windmills to pump water and later to generate electricity

http://www1.eere.energy.gov/windandhydro/printable_versions/wind_history.html

http://www.cutyourfootprint.com

http://thebeliever07.wordpress.com

HISTORY OF WIND ENERGY

- Industrialization, first led to decline in the use of windmills. The steam engine replaced European water-pumping windmills. However, industrialization also sparked the development of larger windmills to generate electricity. Commonly called wind turbines, these machines appeared in Denmark as early as 1890.
- In the 1940s the largest wind turbine of the time began operating on a Vermont hilltop known as Grandpa's Knob. This turbine, rated at 1.25 megawatts in winds of about 30 mph, fed electric power to the local utility network for several months during World War II.

http://www.azsolarcenter.com/imagesets/series01.html

http://www1.eere.energy.gov/windandhydro/printable_versions/wind_history.html

HISTORY OF WIND ENERGY

- The popularity of using the energy in the wind has always fluctuated with the price of fossil fuels. When fuel prices fell after World War II, interest in wind turbines waned. But when the price of oil skyrocketed in the 1970s, so did worldwide interest in wind turbine generators.
- Today, wind energy is the world's fastest-growing energy source and will power industry, businesses and homes with clean, renewable electricity for many years to come.

http://green.venturebeat.com/2009/10/14/siemens-closer-to-wind-dominance-with-900m-in-turbine-contracts/

http://www1.eere.energy.gov/windandhydro/printable_versions/wind_history.html

WIND POWER

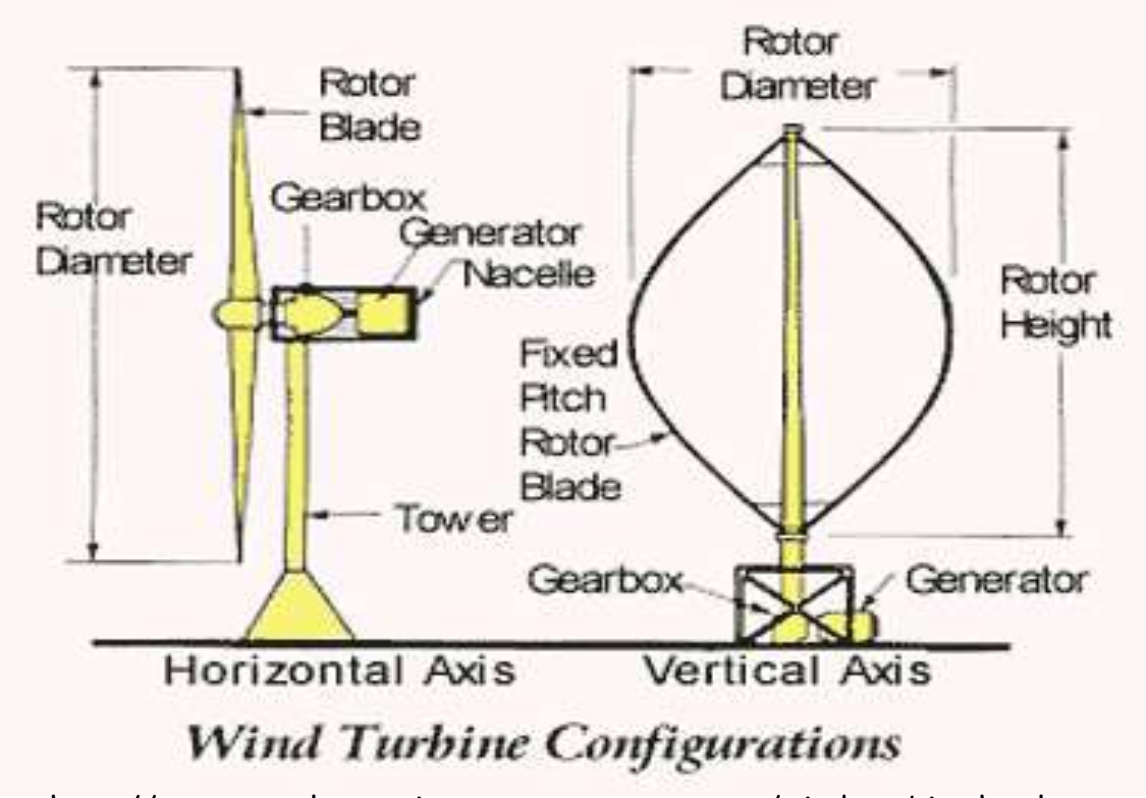
- Wind turbines convert the kinetic energy in the wind into mechanical power.
- This mechanical power is converted into electricity using a generator (similar to hydropower)
- There two types of wind turbine:
  - Horizontal axis (more common)
  - Vertical axis

http://www.solarnavigator.net/wind_turbines.htm

http://www.worldofenergy.com.au/factsheet_wind/07_fact_wind_WA.html

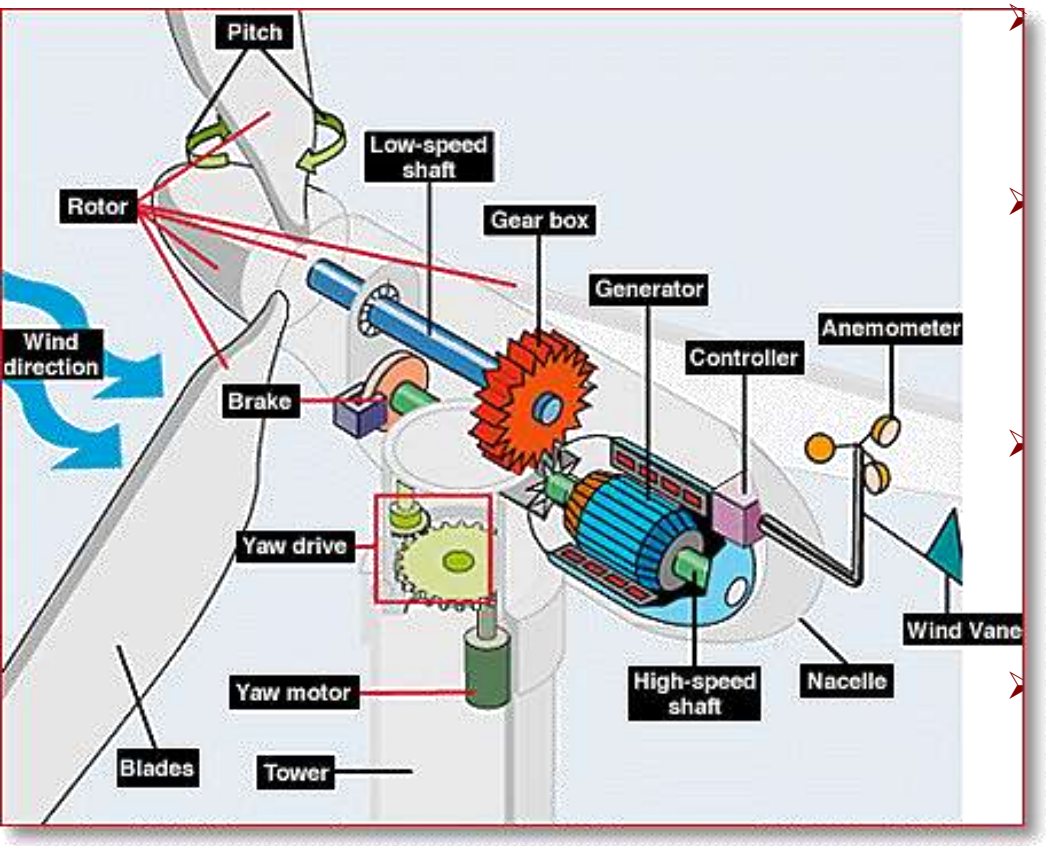
HOW IT WORKS?

- Similar to hydropower, opposite of fan
- The wind turns the blades, the blades spin the shaft connected to a generator that generates electricity
- The rotor and the generator are in a tower in horizontal axis systems to capture more wind energy
- The generator is on the land in vertical system



http://www.top-alternative-energy-sources.com/wind-turbine.html

BASIC ELEMENTS



http://www1.eere.energy.gov/windandhydro

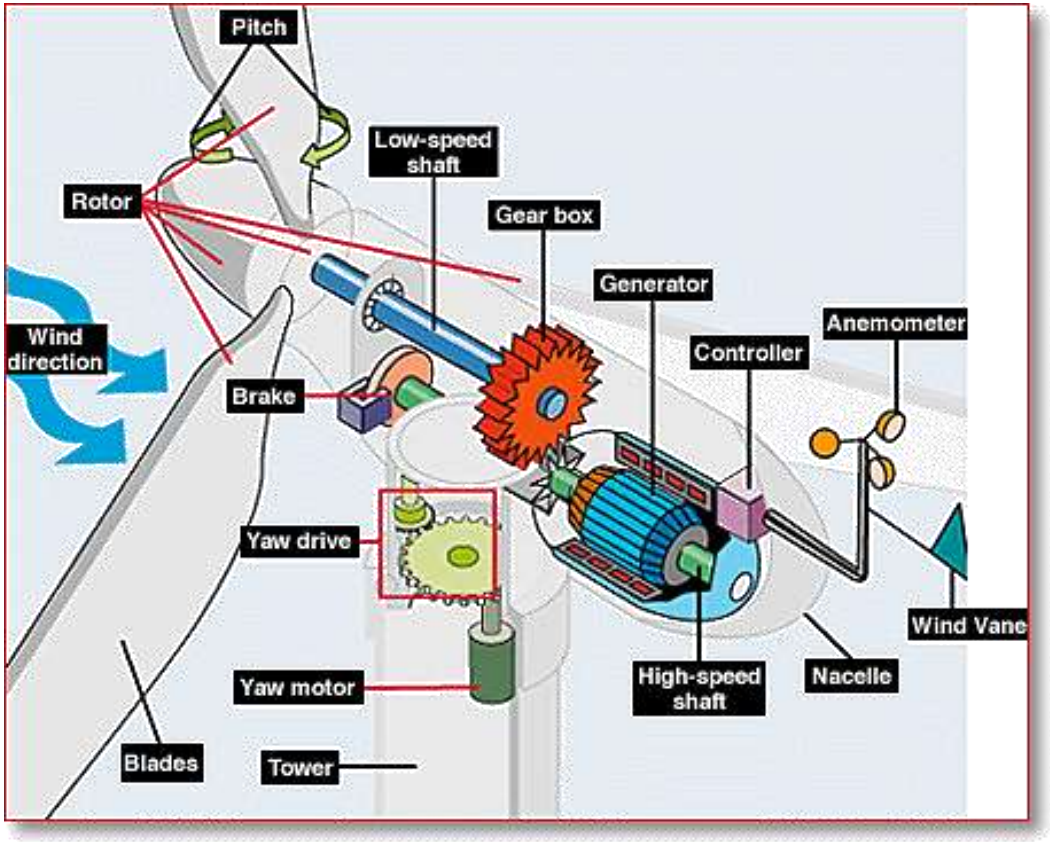
Anemometer: Measures the wind speed and transmits wind speed data to the controller.

Blades: Most turbines have either two or three blades. Wind blowing over the blades causes the blades to "lift" and rotate.

Brake: A disc brake, which can be applied mechanically, electrically, or hydraulically to stop the rotor in emergencies.

Controller: The controller starts up the machine at wind speeds of about 8 to 16 miles per hour (mph) and shuts off the machine at about 55 mph. Turbines do not operate at because they might be damaged

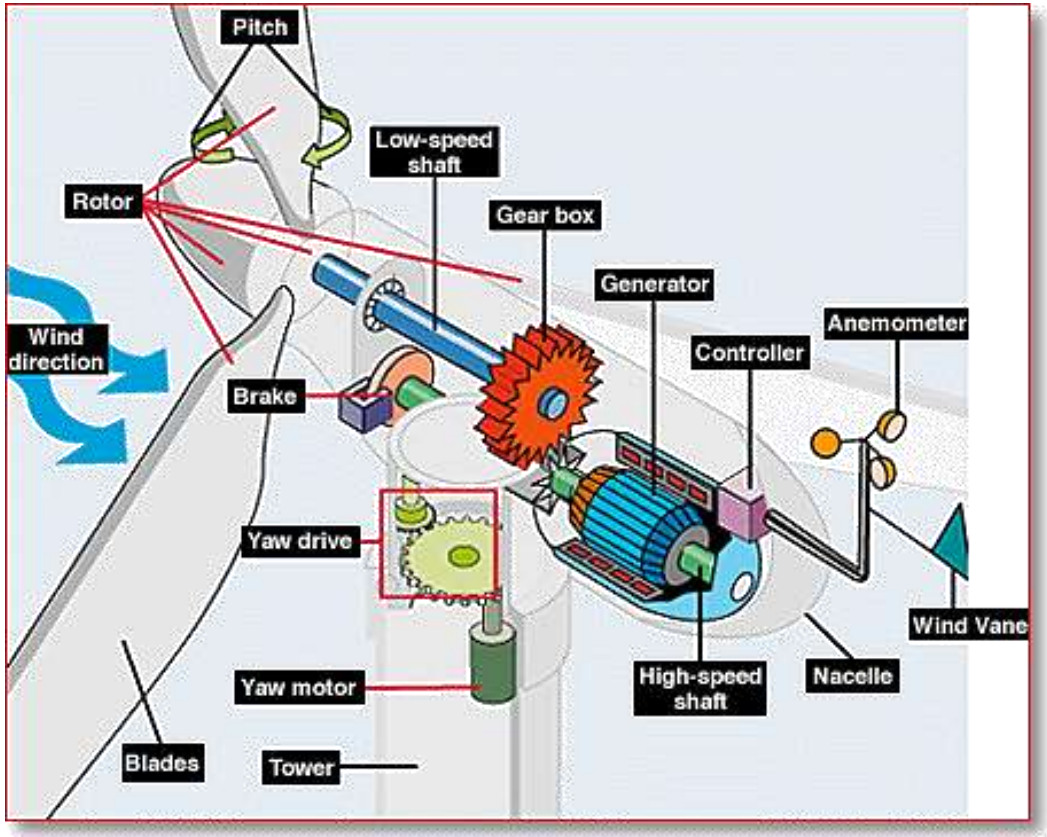
BASIC ELEMENTS



- **High-speed shaft:** Drives the generator.
- **Low-speed shaft:** The rotor turns the low-speed shaft
- **Gear box:** Gears connect the low-speed shaft to the high-speed shaft and increase the rotational speeds from about 30 to 60 rotations per minute (rpm) to about 1000 to 1800 rpm,
- **Generator:** Usually an off-the-shelf induction generator that produces 60-cycle AC electricity.

http://www1.eere.energy.gov/windandhydro

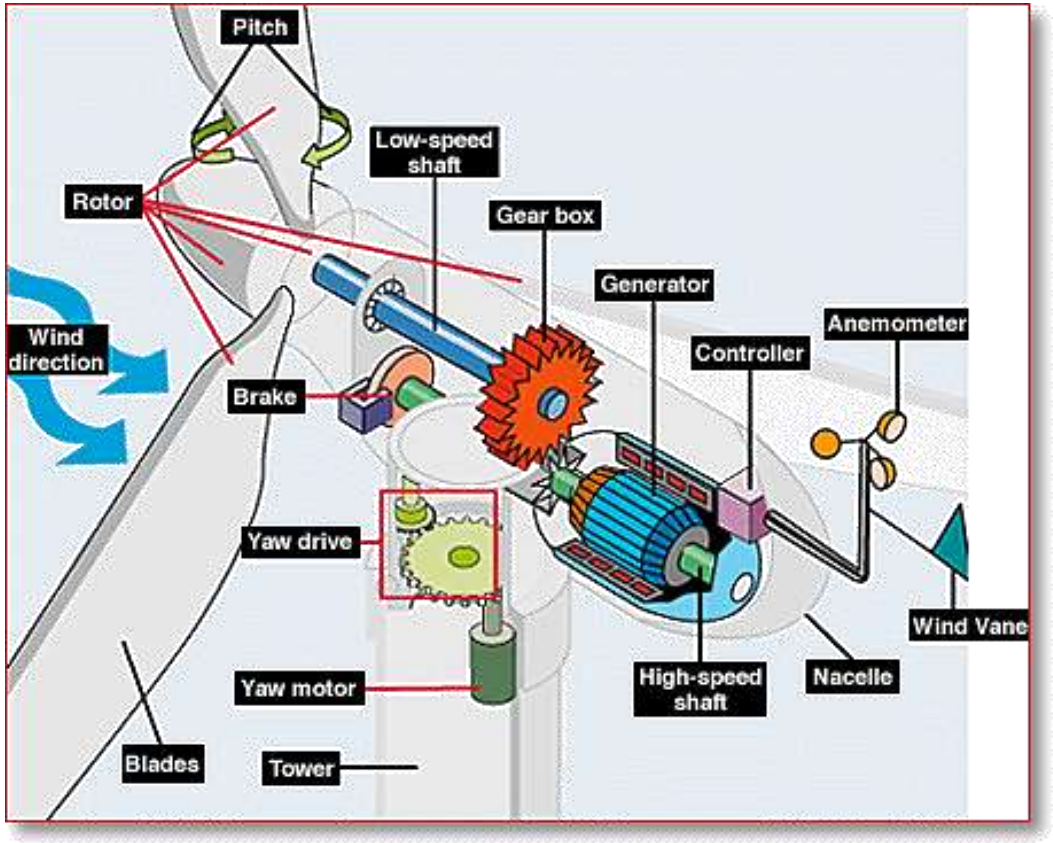
BASIC ELEMENTS



- **Nacelle:** The nacelle contains the gear box, low- and high-speed shafts, generator, controller, and brake.
- **Pitch:** Blades are pitched out of wind to control rotor speed and keep rotor from turning in winds that are too high or too low
- **Rotor:** The blades and the hub together are called the rotor.
- **Tower:** Towers are made from tubular steel, concrete, or steel lattice.

http://www1.eere.energy.gov/windandhydro

BASIC ELEMENTS



http://www1.eere.energy.gov/windandhydro

- **Wind vane:** Measures wind direction and communicates with the yaw drive to orient the turbine with respect to the wind.
- **Yaw drive:** "Upwind" turbines face into the wind; the yaw drive is used to keep the rotor facing into the wind as the wind direction changes. Downwind turbines (facing away wind) don't require a yaw drive, the wind blows the rotor downwind.
- **Yaw motor:** Powers the yaw drive.

BASIC ELEMENTS

Table 3. Percentage of Materials Used in Current Wind Turbine Component

	Large Turbines and *(Small Turbines*[1])							
Component/ Material (% by weight)	Permanent Magnetic Materials	Pre-stressed Concrete	Steel	Aluminum	Copper	Glass Reinforced Plastic [4]	Wood Epoxy [4]	Carbon Filament Reinforced Plastic [4]
Rotor								
Hub			*(95)* - 100	*(5)*				
Blades			5			95	*(95)*	*(95)*
Nacelle [2]	*(17)*		*(65)* - 80	3 - 4	14	1 - *(2)*		
Gearbox [3]			98 -*(100)*	*(0)* - 2	*(<1)* - 2			
Generator	*(50)*		*(20)* - 65		*(30)* - 35			
Frame, Machinery & Shell			85 - *(74)*	9 - *(50)*	4 - *(12)*	3 - *(5)*		
Tower		2	98	*(2)*				

Notes:
1. Small turbines with rated power less than 100 kW- (listed in italics where different)
2. Assumes nacelle is 1/3 gearbox, 1/3 generator and 1/3 frame & machinery
3. Approximately half of the small turbine market (measured in MW) is direct drive with no gearbox
4. Rotor blades are either glass reinforced plastic, wood-epoxy or injection molded plastic with carbon fibers

http://www.perihq.com/documents/WindTurbine-MaterialsandManufacturing_FactSheet.pdf

POWER OF WIND

Rate of kinetic energy of wind entering the cross section of rotor:

$Ek = \frac{1}{2} mv^2$

Power (P) =energy/time

$P=E_k/t = (\frac{1}{2}mv^2)/t$

$m=\rho V= \rho AL; \quad A=\Pi R^2$

$P= \frac{1}{2} (\rho ALv^2)/t= \frac{1}{2} (\rho Av^2) (L/t)$ $\dot{m}$

$P= \frac{1}{2}\rho Av^3 = \frac{1}{2}\rho\Pi R^2 v^3$

- Then power depends on the wind speed, blade diameter, density of air (for example temperature)
- But not all of this power can be converted to electricity

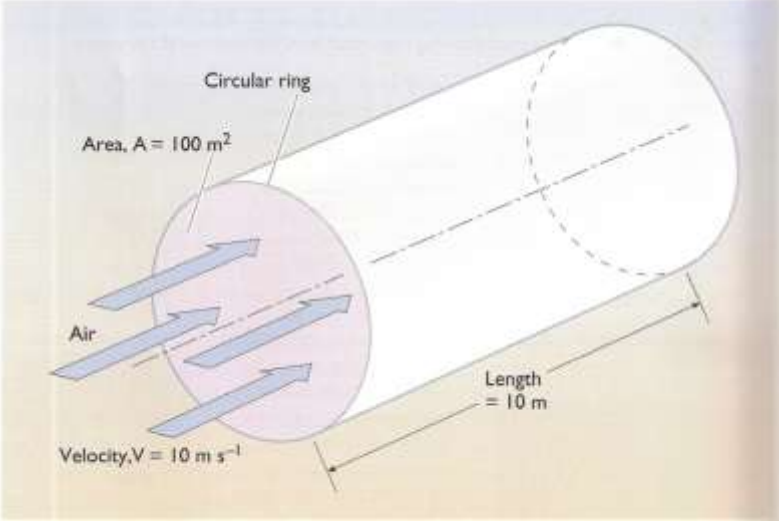


Figure 7.7 Cylindrical volume of air passing at velocity V (10 m s⁻¹) through a ring enclosing an area, A, each second

Boyle, Renewable Energy, Oxford, 2004

Alternatively:

$P=(\frac{1}{2}mv^2)/t= (\frac{1}{2}v^2)(m/t)$

$P= (\frac{1}{2}v^2)(\rho V/t)$

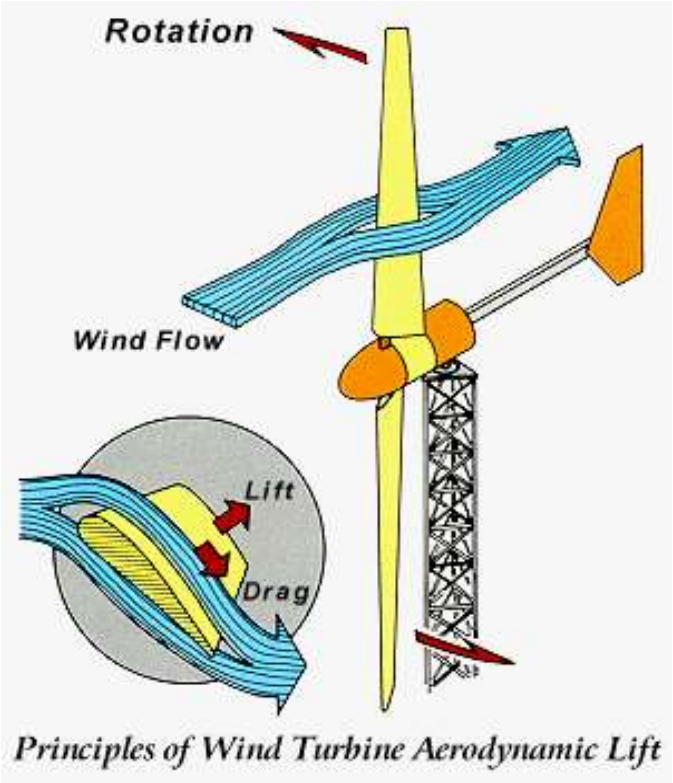
$P= (\frac{1}{2}v^2)(\rho Q)$

$P= (\frac{1}{2}v^2)(\rho Av)$

$P= \frac{1}{2}\rho Av^3 = \frac{1}{2}\rho\Pi R^2 v^3$

BLADE STRUCTURE

- The wind passes more rapidly over the longer (upper) side of the airfoil, creating a lower- pressure area above the airfoil.
- The pressure differential between top and bottom surfaces results in **"lift"** force causing rotation of blade
- In addition, a **"drag"** force perpendicular to the lift force impedes rotor rotation.
- A prime objective in wind turbine design is for the blade to have a relatively high lift-to-drag ratio.



http://www.awea.org/faq/basicop.html

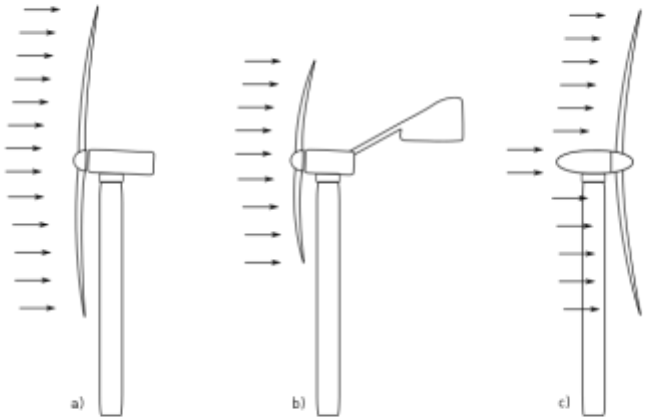
UPWIND/DOWNWIND TURBINES

Upwind Turbines

- Avoid wind shading of tower
- Need yawn mechanism
- Rotor needs to be inflexible and away from the tower

Downwind Turbines

- No need for yawn mechanism (for small turbines)
- Rotor may be more flexible
- Wind shading of tower and fatigue due to the non-uniform wind



http://en.wikipedia.org/wiki/Yaw_system

Upwind turbines are more common

NUMBER OF BLADES

Rotor Solidity

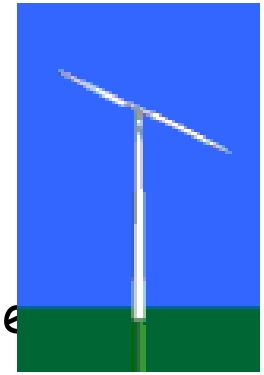
The ratio of blade area to total swept area

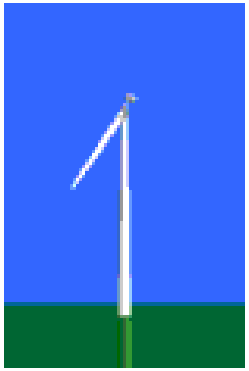
High Solidity: Low speed, high torque. Good for water pumping type of activities

Low Solidity: High speed, low torque. Good for electricity production. Lower risk of damage under very high wind speed or turbulence

Odd/even number of blades: Odd numbers are more stable (consider symmetry and load differences in the up and down position of blade

Three blade rotors are common (there are also one or two blade turbines due to their lower cost)





ROTOR SIZE

- Could be in variety of sizes depending on the use

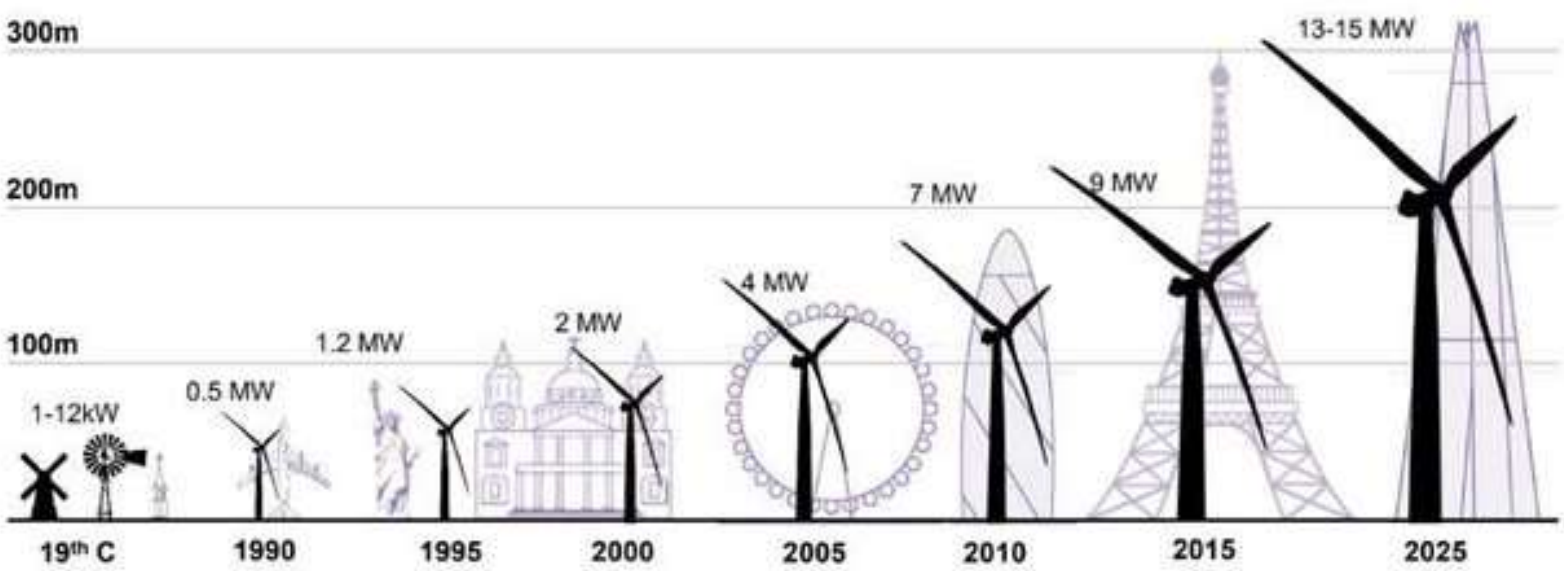
http://www.medwayports.com/news/08.htm

http://www.groovygreen.com/groove/?cat=29

DOI: 10.13140/RG.2.2.33222.93769

ROTOR SIZE

- Larger the rotor size, much larger the energy captured (Remember $P = \frac{1}{2}\rho \Pi R^2 v^3$)
- Larger the rotor size, stronger the wind you need (there could be longer down times in low wind locations and less power generation compared to smaller turbines)



Enercon 126

Blade Diameter: 126 m

Capacity: 7 MW (enough for about 5000 household in Europe)

Build by Enercon, Germany

ROTOR SIZE

Small (≤10 kW)

- Homes (Grid connected)
- Farms
- Remote Applications

(e.g. battery changing, water pumping, telecom sites, icemaking)

Intermediate (10-500 kW)

- Village Power
- Hybrid Systems
- Distributed Power

Large (500 kW – 6 MW)

- Central Station Wind Farms
- Distributed Power
- Offshore Wind Generation Stations

Brian Parsons, NREL (http://www1.eere.energy.gov)

ROTOR SIZE

Reasons for Choosing Large Turbines

- Economy of scale, i.e. larger machines are usually deliver electricity at a lower cost (The cost of foundations, road, electrical grid connection, and some components are independent of the size of the machine.
- Larger machines are well suited for offshore wind power. The cost of foundations is not proportional to the size and maintenance costs are independent of the size.
- In areas where it is difficult to find sites for more than a single turbine, a large turbine with a tall tower uses the wind resource more efficiently.

ROTOR SIZE

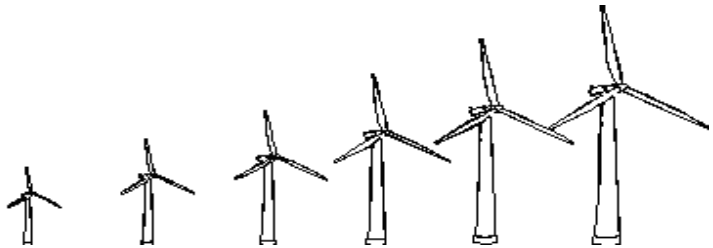
Reasons for Choosing Smaller Turbines

- The local grid may be too weak to handle the output of large turbine
- Less fluctuation in the output from a wind park consisting of a number of smaller machines (fluctuations tend to cancel out)
- The cost of using large cranes and building roads to carry the turbine components may make smaller turbines more economical in some areas.
- Several smaller machines spread the risk of temporary machine failure, e.g. due to lightning strikes.
- Aesthetical landscape considerations may sometimes dictate the use of smaller machines (but one single large turbine could be better than many small turbine in some cases).

TOWER HEIGHT

- High tower for larger rotor size
- Higher tower means larger wind power but also higher costs
- Tower height will be depend on the wind characteristics of location and economics
- Usually equal to rotor diameter (more aesthetic)

Enercon E33
Rated power: 330 kW
Rotor diameter: 33.4 m
Hub height: 37 m – 50 m



LOCATION

Location can effects the performance of the wind turbine in many ways:

- Wind speed distribution
- Roughness, obstacles and tunnels, valleys hills,
- Closeness to the living area (noise, aesthetics, interference ...)
- Closeness to the grid connection
- ....

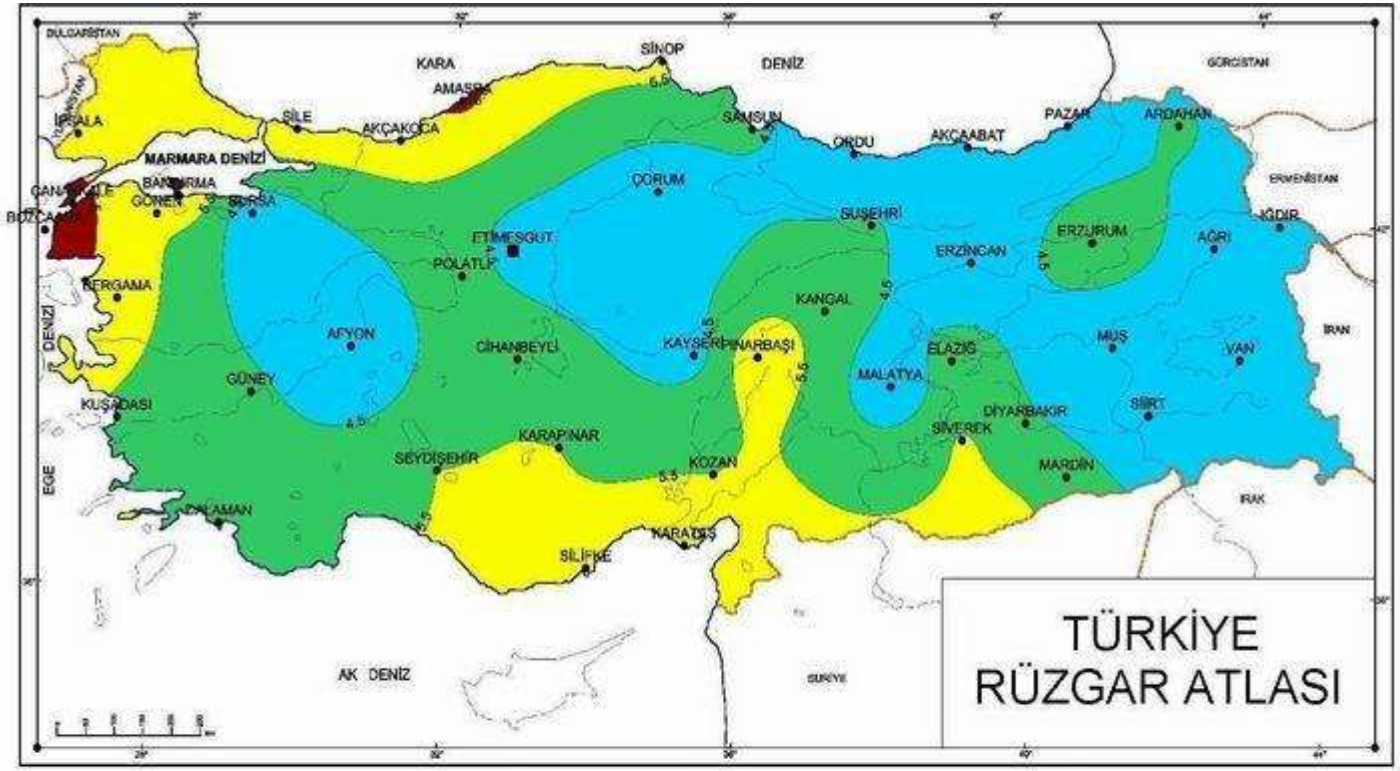
LOCATION: Wind speed

- Usually 4 or higher class winds were preferred for large scale operations

Classes of Wind Power Density at 10 m and 50 m[a]				
	10 m (33 ft)		50 m (164 ft)	
Wind Power Class	Wind Power Density (W/m^2)	Speed[b] m/s (mph)	Wind Power Density (W/m^2)	Speed[b] m/s (mph)
1	<100	<4.4 (9.8)	<200	<5.6 (12.5)
2	100 - 150	4.4 (9.8)/5.1 (11.5)	200 - 300	5.6 (12.5)/6.4 (14.3)
3	150 - 200	5.1 (11.5)/5.6 (12.5)	300 - 400	6.4 (14.3)/7.0 (15.7)
4	200 - 250	5.6 (12.5)/6.0 (13.4)	400 - 500	7.0 (15.7)/7.5 (16.8)
5	250 - 300	6.0 (13.4)/6.4 (14.3)	500 - 600	7.5 (16.8)/8.0 (17.9)
6	300 - 400	6.4 (14.3)/7.0 (15.7)	600 - 800	8.0 (17.9)/8.8 (19.7)
7	>400	>7.0 (15.7)	>800	>8.8 (19.7)

http://www.awea.org/faq/basicwr.html

LOCATION



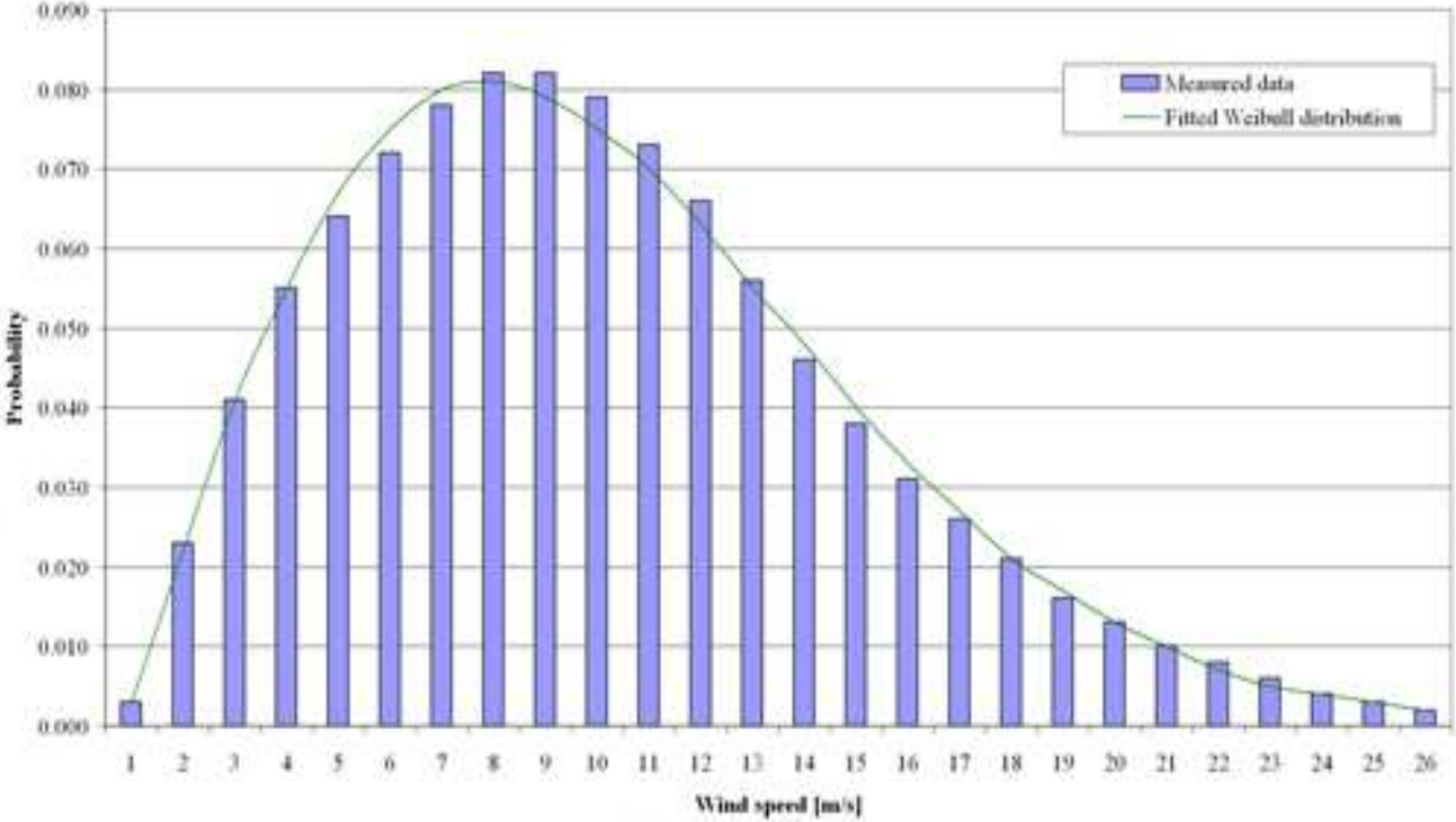
U (m/s)	> 7.5	6.5 – 7.5	5.5 –6.5	4.5 – 5.5	< 4.5
P (W / m^2)	> 500	300 - 500	200 - 300	100 - 200	< 100

* *Açık yüzeyler için (yer düzeyinden 50 m yükseklikteki) rüzgar potansiyeli sınıf aralıkları*

http://www.meteoroloji.gov.tr

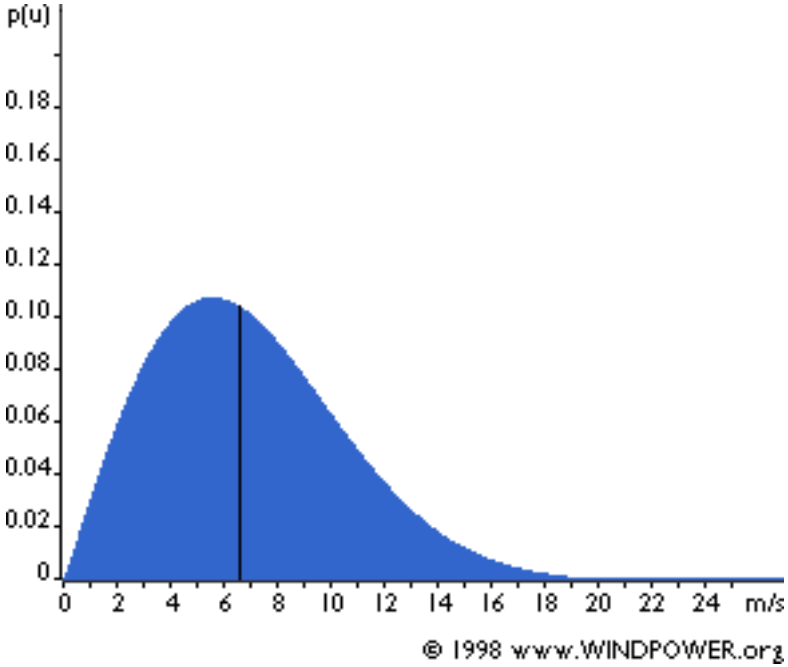
LOCATION: Wind distrubution

- Wind speed in a location is not constant



LOCATION: Wind distrubution

- It can be modeled using the **Weibull** Distribution



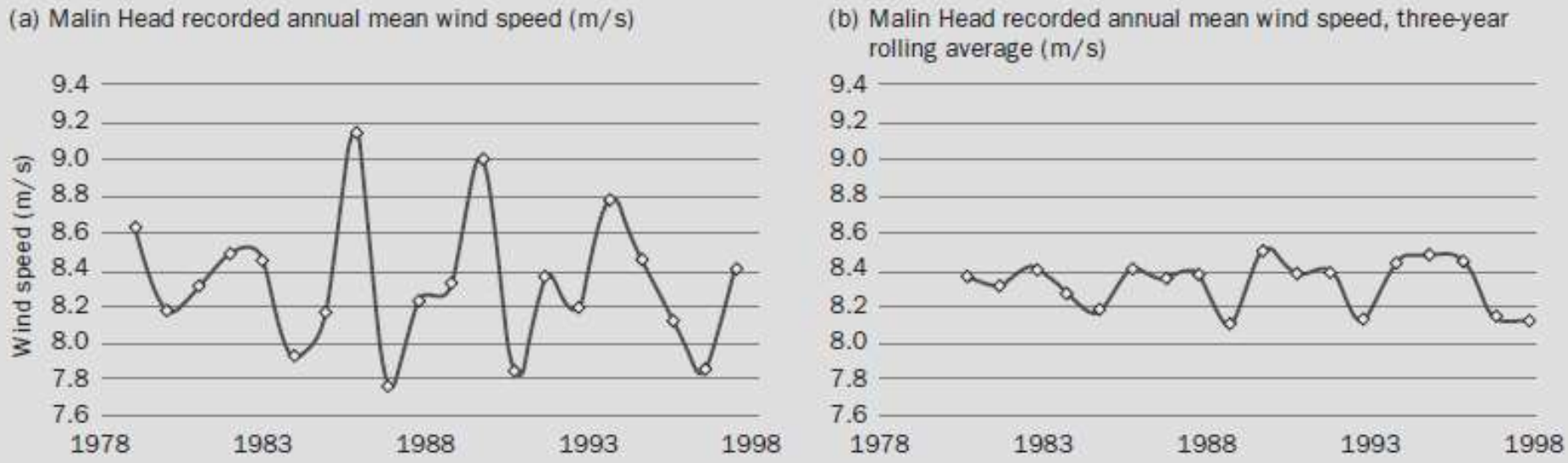
$$f(x;\lambda,k) = \begin{cases} \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

k: shape factor
λ: scale factor

If k=2, we get Rayleigh distribution, which represent the wind distribution best in most cases

LOCATION: Wind distrubution

Figure I.2.8: (a) The annual mean wind speed recorded at Malin Head, Ireland; (b) annual mean wind speed at Malin Head over a 20-year period – three-year rolling averages



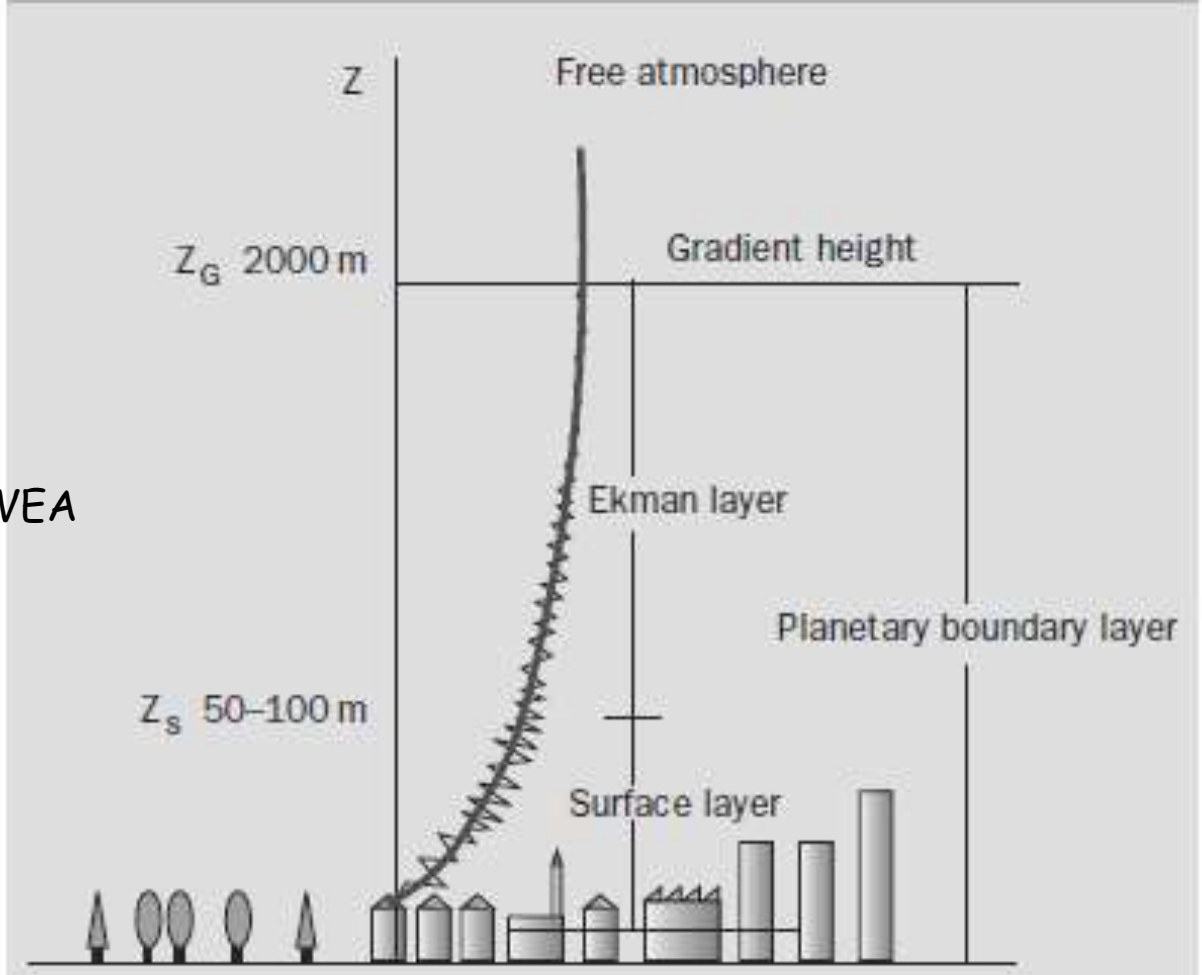
Source: Garrad Hassan

The III. Wind Energy Facts I, EWEA

LOCATION: Roughness

The Wind Energy Facts I, EWEA

Figure I.2.3: The atmospheric boundary layer shear profile



Source: Garrad Hassan

LOCATION: Roughness

$$v = v_{ref} \frac{\ln(z / z_o)}{\ln(z_{ref} / z_o)}$$

v: speed at height z

v_{ref}: spped at heigh z_{ref}

z_o: roughness

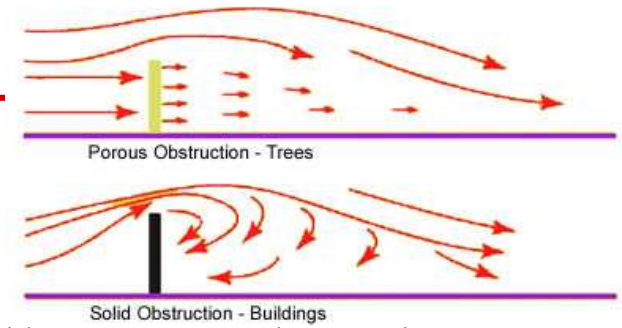
http://www.talentfactory.dk
(Danish Wind Industry Association)

Roughness Classes and Roughness Length Table

Rough-ness Class	Roughness Length m	Energy Index (per cent)	Landscape Type
0	0.0002	100	Water surface
0.5	0.0024	73	Completely open terrain with a smooth surface, e.g.concrete runways in airports, mowed grass, etc.
1	0.03	52	Open agricultural area without fences and hedgerows and very scattered buildings. Only softly rounded hills
1.5	0.055	45	Agricultural land with some houses and 8 metre tall sheltering hedgerows with a distance of approx. 1250 metres
2	0.1	39	Agricultural land with some houses and 8 metre tall sheltering hedgerows with a distance of approx. 500 metres
2.5	0.2	31	Agricultural land with many houses, shrubs and plants, or 8 metre tall sheltering hedgerows with a distance of approx. 250 metres
3	0.4	24	Villages, small towns, agricultural land with many or tall sheltering hedgerows, forests and very rough and uneven terrain
3.5	0.8	18	Larger cities with tall buildings
4	1.6	13	Very large cities with tall buildings and skycrapers

LOCATION: Wind obstacles

- Obstacles (trees, houses, etc) prevent or decreases the wind, and/or creates turbulence



http://www.kaper.us/basics/BASICS_040502_hunt_windflow.html

Wind Speed in per cent of Wind Speed Without Obstacle

m height	21	43	64	86	107	129	150	171	193	214	236	257	279	300	321	343	364	386	407	429	450 m
75	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	100	99	99	99	99
73	100	100	100	100	100	100	100	100	100	100	100	100	100	100	99	99	99	99	99	99	99
70	100	100	100	100	100	100	100	100	100	100	100	99	99	99	99	99	99	99	99	99	99
68	100	100	100	100	100	100	100	100	100	99	99	99	99	99	99	99	99	99	99	99	99
65	100	100	100	100	100	100	100	100	99	99	99	99	99	99	99	99	99	99	99	99	99
63	100	100	100	100	100	100	100	99	99	99	99	99	99	99	99	99	99	99	99	99	99
60	100	100	100	100	100	100	100	99	99	99	99	99	99	98	98	98	98	98	98	98	98
58	100	100	100	100	100	99	99	99	99	98	98	98	98	98	98	98	98	98	98	98	98
55	100	100	100	100	100	99	99	98	98	98	98	98	98	98	98	98	98	98	98	98	98
53	100	100	100	100	99	99	98	98	98	97	97	97	97	97	97	97	97	97	98	98	98
50	100	100	100	99	99	98	98	97	97	97	97	97	97	97	97	97	97	97	97	97	97
48	100	100	100	99	98	98	97	96	96	96	96	96	96	96	96	96	96	97	97	97	97
45	100	100	99	99	98	97	96	95	95	95	95	95	95	95	95	96	96	96	96	97	97
43	100	100	99	98	96	95	94	94	93	93	94	94	94	94	95	95	95	96	96	96	96
40	100	100	98	96	95	93	93	92	92	92	92	93	93	94	94	94	95	95	95	96	96
38	100	99	97	95	93	91	90	90	90	90	91	91	92	93	93	94	94	95	95	95	96
35	100	98	95	92	90	88	88	87	87	88	89	90	91	92	92	93	94	94	94	95	95
33	100	97	92	88	86	85	84	84	85	86	87	89	90	91	91	92	93	93	94	94	95
30	99	94	87	83	81	80	81	81	82	84	86	87	88	90	91	92	92	93	94	94	95
28	99	89	81	76	75	75	76	78	79	81	84	86	87	89	90	91	92	93	93	94	94
25	96	81	71	68	68	70	72	74	76	79	82	84	86	88	89	90	91	92	93	94	94
23	90	68	59	58	60	63	67	70	73	77	80	83	85	87	89	90	91	92	93	93	94
20	78	48	43	46	52	57	63	67	71	75	79	82	85	87	88	90	91	92	93	94	94
18	52	21	24	34	43	52	59	65	69	74	79	82	85	87	88	90	91	92	93	94	94
15	4		4	22	36	48	56	63	69	74	79	82	85	87	89	90	91	92	93	94	95
13				13	32	46	56	63	69	75	79	83	86	88	90	91	92	93	94	94	95
10				9	32	47	58	65	71	77	81	85	87	89	91	92	93	94	95	95	96
8				14	38	53	63	70	75	80	84	87	89	91	92	93	94	95	96	96	97
5				31	51	63	72	78	82	85	88	91	92	94	94	95	96	96	97	97	98
3			36	60	72	80	84	88	90	92	94	95	96	97	97	98	98	98	98	99	99

= Obstacle 20 m tall = Turbine Tower 50 m tall

Roughness length = 0.055; Porosity = 0; Obstacle length = 60 m

Note: Vertical and horizontal scales are different. Horizontal scale shows distance from obstacle

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http://www.talentfactory.dk

(Danish Wind Industry Association)

Wind Energy in per cent of Wind Energy Without Obstacle

m height	21	43	64	86	107	129	150	171	193	214	236	257	279	300	321	343	364	386	407	429	450 m
75	100	100	100	100	100	100	100	100	100	99	99	99	99	99	99	99	99	98	98	98	98
73	100	100	100	100	100	100	100	100	99	99	99	99	99	99	98	98	98	98	98	98	98
70	100	100	100	100	100	100	100	100	99	99	99	99	98	98	98	98	98	98	98	98	98
68	100	100	100	100	100	100	100	99	99	99	98	98	98	98	98	97	97	97	97	97	97
65	100	100	100	100	100	100	99	99	98	98	98	97	97	97	97	97	97	97	97	97	97
63	100	100	100	100	100	99	99	98	98	97	97	97	96	96	96	96	96	96	96	96	96
60	100	100	100	100	99	99	98	98	97	96	96	96	96	95	95	95	95	95	95	95	95
58	100	100	100	100	99	98	98	97	96	95	95	95	95	94	94	94	94	94	95	95	95
55	100	100	100	99	99	98	97	96	95	94	94	93	93	93	93	93	93	94	94	94	94
53	100	100	100	99	98	97	95	94	93	92	92	92	92	92	92	92	92	93	93	93	93
50	100	100	100	98	97	95	93	92	91	90	90	90	90	90	91	91	91	91	92	92	92
48	100	100	99	97	95	93	91	89	88	88	88	88	88	88	89	89	90	90	91	91	91
45	100	100	98	96	93	90	88	86	85	85	85	85	86	87	87	88	88	89	89	90	90
43	100	99	97	93	90	86	84	82	81	81	82	83	84	84	85	86	87	87	88	89	89
40	100	99	95	90	85	82	79	78	77	77	79	80	81	82	83	84	85	86	87	88	88
38	100	98	91	85	79	76	74	72	72	73	75	76	78	80	81	82	83	85	86	86	87
35	100	95	86	78	72	69	67	67	67	68	71	73	75	77	79	80	82	83	84	85	86
33	99	91	78	69	63	61	60	60	61	64	67	70	72	74	77	78	80	82	83	84	85
30	98	83	67	57	53	52	52	53	55	59	63	66	69	72	75	77	79	80	82	83	85
28	96	71	53	45	42	43	44	47	50	54	59	63	67	70	73	75	77	79	81	83	84
25	89	53	36	31	31	34	37	40	44	50	55	60	64	68	71	74	76	78	80	82	84
23	74	31	20	19	22	26	30	35	40	46	52	58	62	66	70	73	76	78	80	82	83
20	47	11	8	10	14	19	25	30	36	43	50	56	61	65	69	72	75	78	80	82	83
18	14	1	1	4	8	14	20	27	33	41	48	55	60	65	69	73	76	78	80	82	84
15				1	5	11	18	25	32	40	48	55	61	66	70	73	76	79	81	83	85
13					3	10	17	25	33	42	50	57	63	68	72	75	78	80	83	84	86
10					3	10	19	28	36	45	54	60	66	71	75	78	80	83	85	86	88
8					5	15	25	34	43	52	60	66	71	75	79	82	84	86	87	89	90
5				3	13	26	37	47	55	62	69	74	78	82	84	87	88	90	91	92	93
3			5	21	37	50	60	67	73	78	82	86	88	90	91	93	94	94	95	96	96

= Obstacle 20 m tall = Turbine Tower 50 m tall

Roughness length = 0.055; Porosity = 0; Obstacle length = 60 m

Note: Vertical and horizontal scales are different. Horizontal scale shows distance from obstacle

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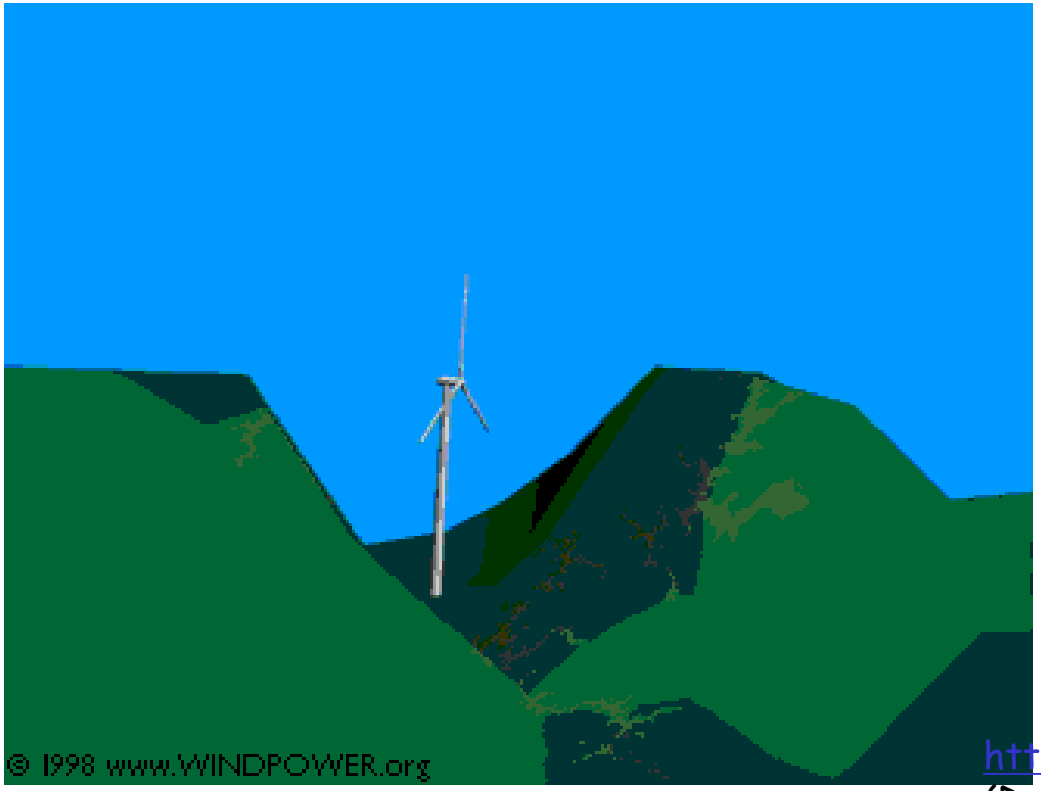
LOCATION:Tunnel effect

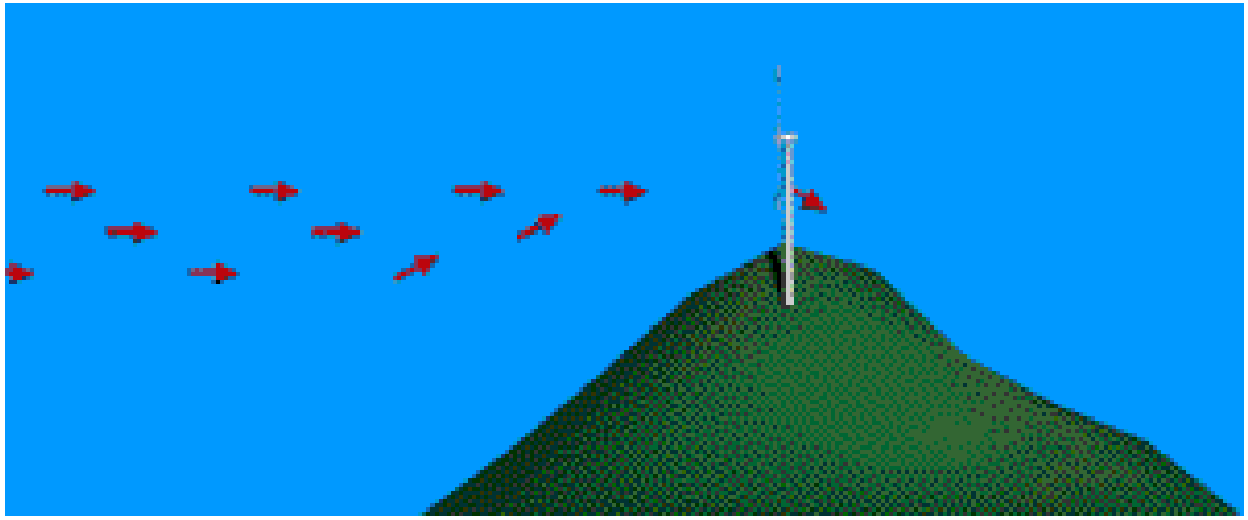
The wind speed, hence the power generated can be increased installing the turbine in a wind tunnel if the turbulence is avoided



http://www.talentfactory.dk
(Danish Wind Industry Association)

LOCATION: Hill effect

Normally the hills are more suitable for wind turbine (higher wind speeds) but the turbulence created behind the hill may have negative effects



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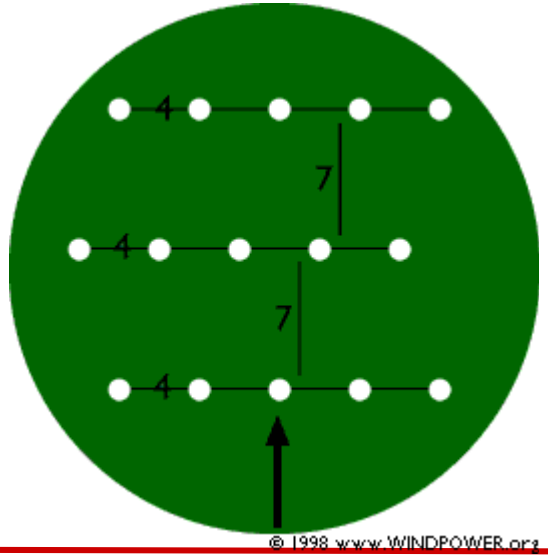
LOCATION: Wake and park effect

- Wind turbine creates wind shade in the downwind direction. there will be a wake (long trail of wind) which is quite turbulent and slowed down
- Turbines in wind parks are usually spaced somewhere between 5 and 9 rotor diameters apart in the prevailing wind direction, and between 3 and 5 diameters apart in the direction perpendicular to the prevailing winds.



http://www.talentfactory.dk
(Danish Wind Industry Association)

ENVIRONMENTA[illegible]e Assessment

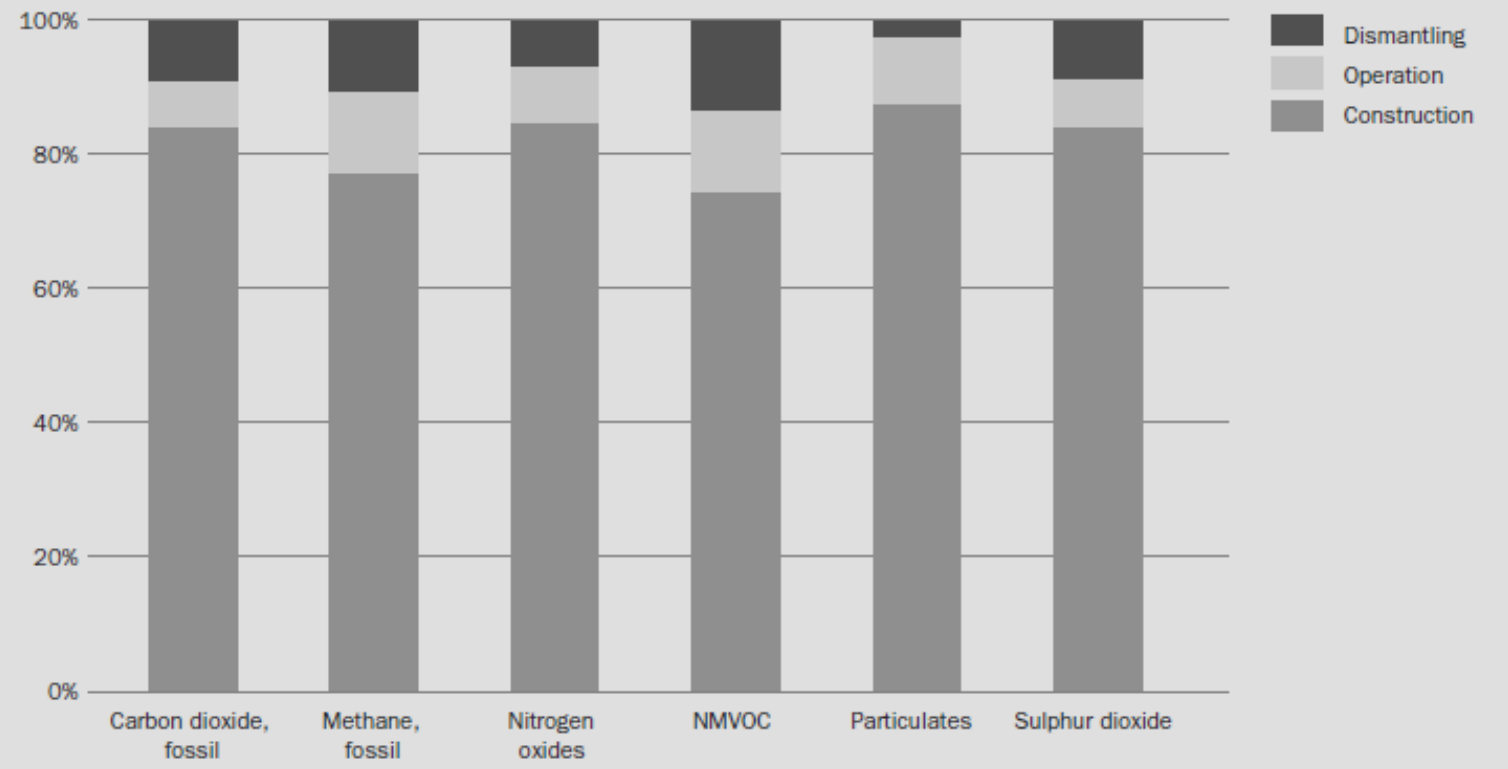
Wind energy LCAs are usually divided into five phases:

1. **Construction** comprises the raw material production (concrete, aluminium, steel, glass fibre and so on) needed to manufacture the tower, nacelle, hub, blades, foundations and grid connection cables.
2. **On-site erection and assembling** includes the work of erecting the wind turbine. This stage used to be included in the construction or transport phases.
3. **Transport** takes into account the transportation systems needed to provide the raw materials to produce the different components of the wind turbine, the transport of turbine components to the wind farm site and transport during operation.
4. **Operation** is related to the maintenance of the turbines, including oil changes, lubrication and transport for maintenance, usually by truck in an onshore scheme.
5. **Dismantling:** once the wind turbine is out of service, the work of dismantling the turbines and the transportation (by truck) from the erection area to the final disposal site; the current scenario includes recycling some components, depositing inert components in landfills and recovering other material such as lubricant oil.

The Wind Energy Facts V,EWEA

ENVIRONMENTAL EFFECTS: Life Cycle Assessment

Figure V.1.3: Contribution of the different life-cycle phases to the relevant emissions



Source: Own elaboration using ECLIPSE results

The Wind Energy Facts V,EWEA

ENVIRONMENTAL EFFECTS: Life Cycle Assessment

Table V.1.3: Emissions and benefits of relevant pollutants produced by wind electricity and other renewable energies

	Emissions					Benefits			
	Average wind	Nuclear	Solar PV	Solar thermal	Biomass CHP	vs. Nuclear	vs. Solar PV	vs. Solar thermal	vs. Biomass CHP
Carbon dioxide, fossil (g)	8	8	53	9	83	0	45	1	75
Methane, fossil (mg)	8	20	100	18	119	12	92	10	111
Nitrogen oxides (mg)	31	32	112	37	814	1	81	6	784
NMVOC (mg)	6	6	20	6	66	0	14	1	60
Particulates (mg)	15	17	107	27	144	1	91	12	128
Sulphur dioxide (mg)	32	46	0	31	250	15	–31	–1	218

Source: CIEMAT

The Wind Energy Facts V,EWEA

ENVIRONMENTAL EFFECTS: Life Cycle Assessment

Table V.1.2: Emissions of relevant pollutants produced by wind electricity and coal and natural gas electricity in the whole life cycle, and benefits of wind versus coal and natural gas

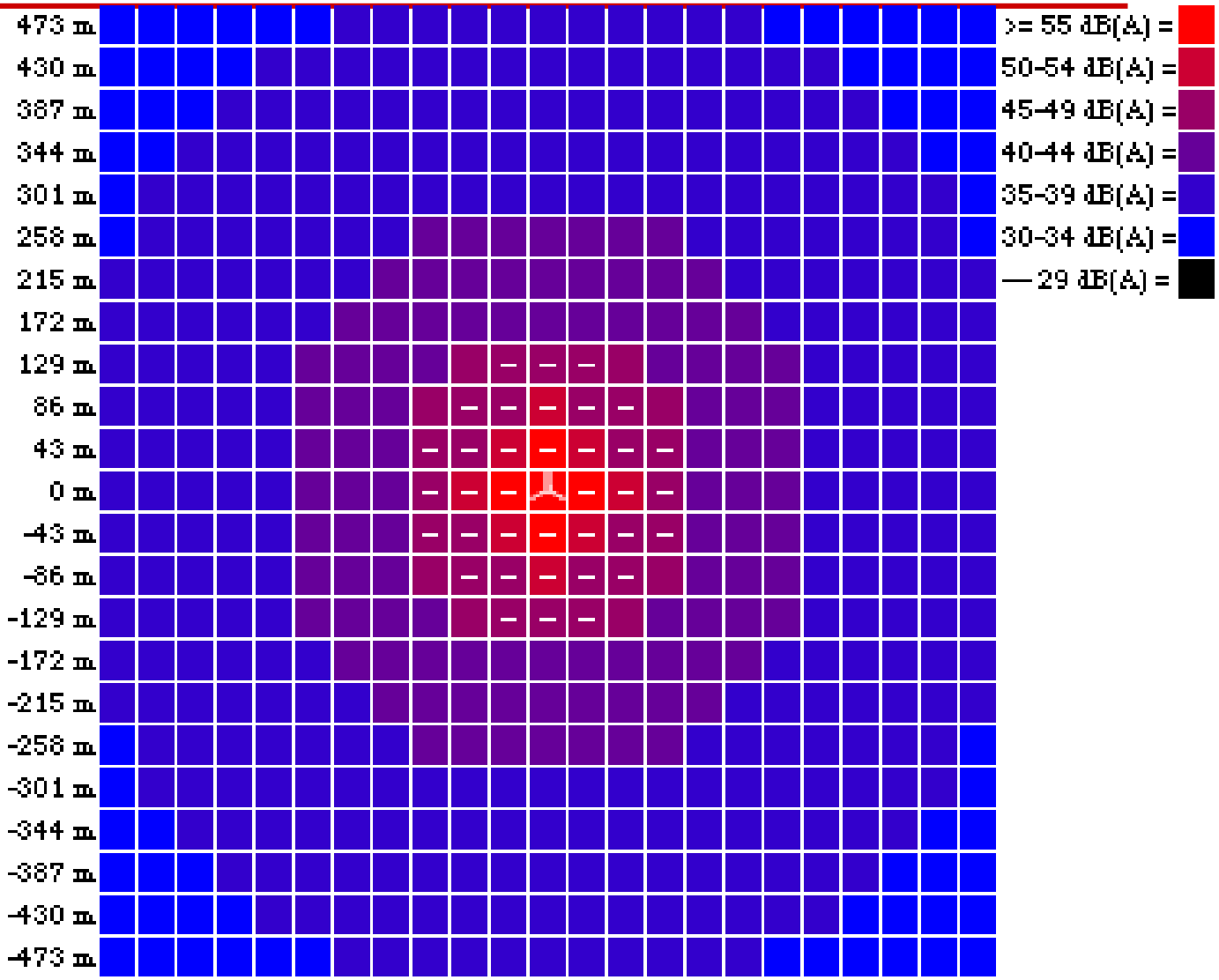
	Emissions						Benefits		
	Onshore wind	Offshore wind	Average wind	Hard coal	Lignite	NGCC	vs. coal	vs. Lignite	vs. NGCC
Carbon dioxide, fossil (g)	8	8	8	836	1060	400	828	1051	391
Methane, fossil (mg)	8	8	8	2554	244	993	2546	236	984
Nitrogen oxides (mg)	31	31	31	1309	1041	353	1278	1010	322
NMVOC (mg)	6	5	6	71	8	129	65	3	123
Particulates (mg)	13	18	15	147	711	12	134	693	–6
Sulphur dioxide (mg)	32	31	32	1548	3808	149	1515	3777	118

Source: CIEMAT

The Wind Energy Facts V,EWEA

ENVIRONMENTAL EFFECTS: Noise

http://www.talentfactory.dk
(Danish Wind Industry Association)



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ENVIRONMENTA

Table V.2.1: Comparative noise for common activities

Source/activity	Indicative noise level (dB)
Threshold of hearing	0
Rural night-time background	20–40
Quiet bedroom	35
Wind farm at 350 m	35–45
Busy road at 5 km	35–45
Car at 65 km/h at 100 m	55
Busy general office	60
Conversation	60
Truck at 50 km/h at 100 m	65
City traffic	90
Pneumatic drill at 7 m	95
Jet aircraft at 250 m	105
Threshold of pain	140

Source: CIEMAT

The Wind Energy Facts V, EWEA

ENVIRONMENTAL EFFECTS: Birds

Table V.2.2: Anthropogenic bird mortality

Causes	Annual mortality estimate
Buildings/windows	550 million
Cats	100 million
High tension lines	130 million
Vehicles	80 million
Pesticides	67 million
Communication towers	4.5 million
Airplanes	25 thousand
Wind turbines	28.5 thousand

Source: Erickson et al. (2005)

The Wind Energy Facts V,EWEA

LOCATION: Environmental concerns

Some other minor effects such

- Shadow
- Arial marking
- Aesthetic
- ....

LOCATION

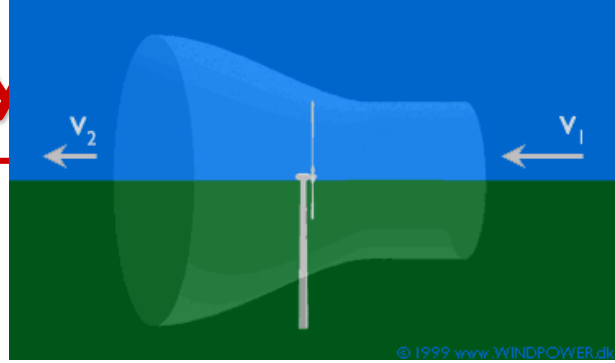
Type of location used

- Onshore
- Off shore (usually 10 km from the shore)
- Near shore

http://glengro.com/wind_energy/Ecology.html

http://www.envirowarrior.com/global-wind-power-stats/

PERFORMANCE AND EFFICIENCY



The Power coefficient: (C_p):

$$C_p = P_t/P$$

P_t: Power extracted by turbine; P available in wind

$$P = \tfrac{1}{2}\rho A v_1^3 = \tfrac{1}{2}\rho \Pi R^2 v_1^3$$

$$P_t = \tfrac{1}{2}(m/t)(v_1^2 - v_2^2)$$

$$P_t = \tfrac{1}{2}(\rho V/t)(v_1^2 - v_2^2)$$

$$P_t = \tfrac{1}{2}(\rho Q)(v_1^2 - v_2^2)$$

$$P_t = \tfrac{1}{2}(\rho A v)(v_1^2 - v_2^2)$$

$$v = (v_1 + v_2)/2$$

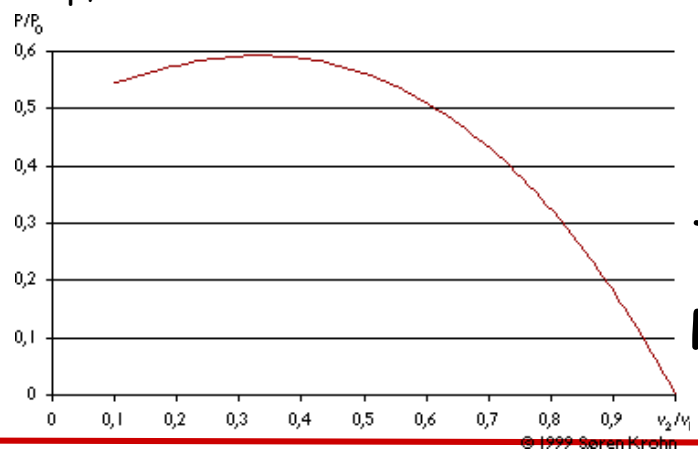
$$P_t = \tfrac{1}{2}\rho A(v_1^2 - v_2^2)(v_1 + v_2)/2$$

$$P_t = \tfrac{1}{4}\rho A(v_1 + v_2)(v_1^2 - v_2^2)$$

$$C_p = P_t/P = \{\tfrac{1}{4}\rho A(v_1 + v_2)(v_1^2 - v_2^2)\}/\{\tfrac{1}{2}\rho A v_1^3\}$$

$$C_p = \tfrac{1}{2}(v_1 + v_2)(v_1^2 - v_2^2)/v_1^3$$

$C_{p,max} = 0.59$ (Betz limit. Theoritical efficiency)



Then:

$$P = 0.59\tfrac{1}{2}\rho \Pi R^2 v^3$$

PERFORMANCE AND EFFICIENCY

The tip speed ratio (λ):

λ= (Tip speed of blade)/(Wind speed)= $(2\Pi r/t)/V$

r: radius of blade; t: time; V: wind speed

There is an optimum value of

$$\lambda_{opt} \cong 4\Pi/n$$

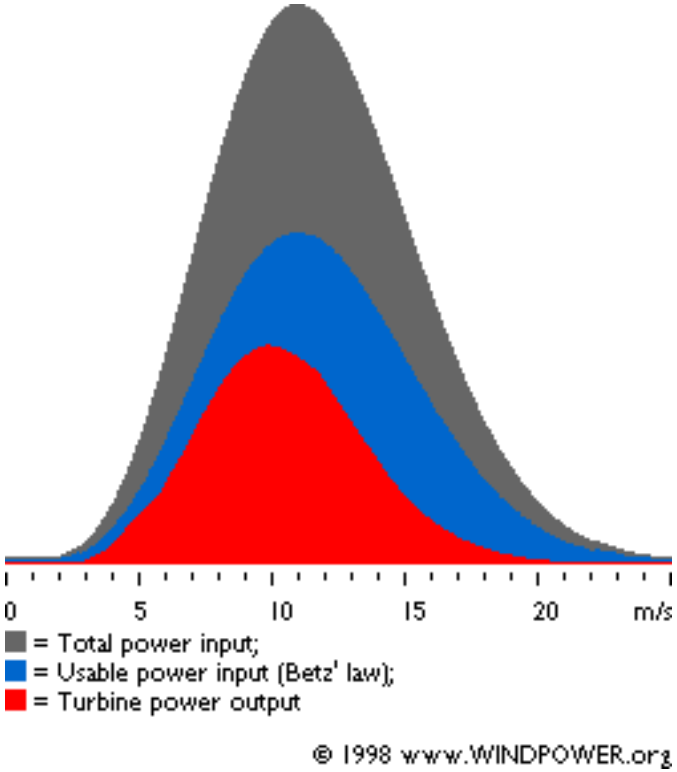
n: # of blade

- If the tip speed ratio is too low (rotor spins too slowly) the wind will pass through the gaps between the blades reducing power extraction
- if the tip speed ratio is too high(rotor spins too fast) the blades will act a solid wall and obstruct the wind again reducing the power extraction

PERFORMANCE AND EFFICIENCY

- Cp is always less than 0.59 in actual conditions
- Cp depends on the tip speed ratio (rotor size and wind speed)
- Since you can optimize Cp for a single value of wind speed, and the wind speed always varies, you can never design a perfectly efficient turbine

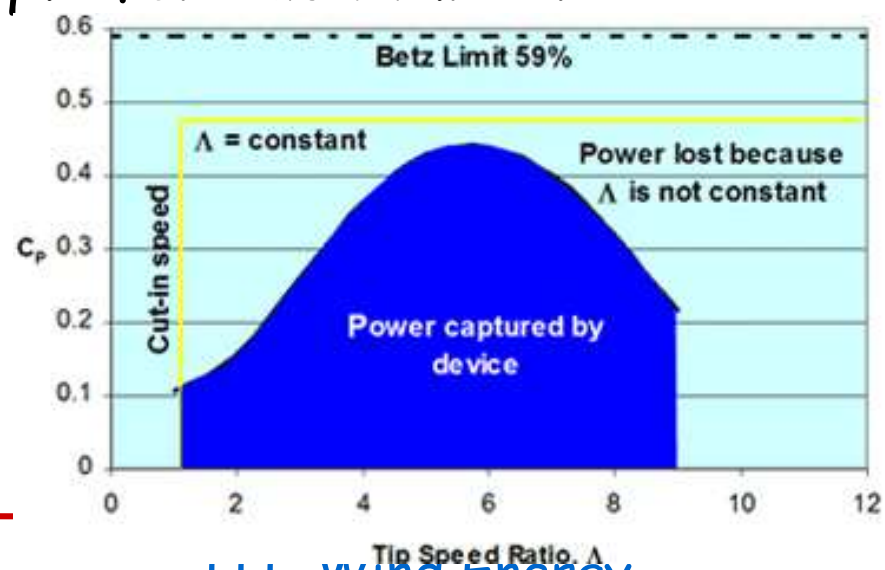
Power density



PERFORMANCE AND EFFICIENCY

- Maximum Cp is usually about 0.4-0.5
- At very low and high wind speed there is no power generation
- <u>Cut-in speed:</u> turbine starts to work
- <u>Cut-out speed:</u> the turbine is stopped (for safety)
- Both of these limits are unique for each turbine

Low efficiency is not important (wind is free), power/$ is the important parameter



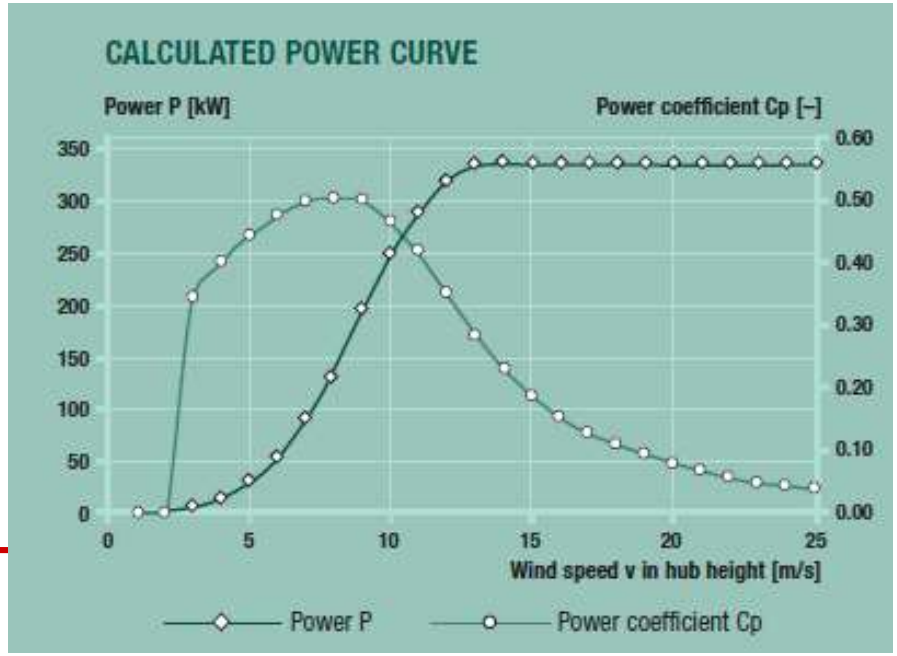
PERFORMANCE AND EFFIC

Enercon E33

TECHNICAL DATA

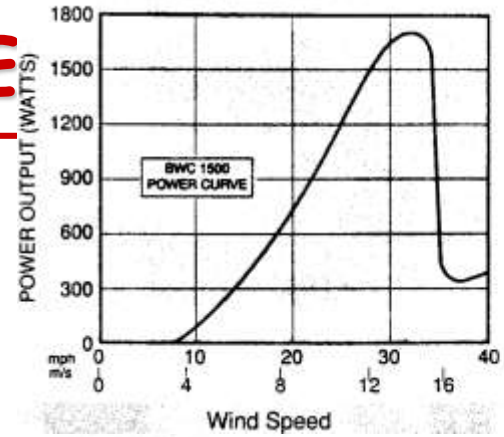
Rated power:	330 kW
Rotor diameter:	33.4 m
Hub height:	37 m – 50 m
Wind class (IEC):	IEC/NVN I and IEC/NVN II (depending on hub height)
Turbine concept:	Gearless, variable speed, variable pitch control
Rotor	
Type:	Upwind rotor with active pitch control
Direction of rotation:	Clockwise
Number of blades:	3
Swept area:	876 m²
Blade material:	Fibreglass (epoxy resin); integrated lightning protection
Rotational speed:	Variable, 18 – 45 rpm
Pitch control:	ENERCON blade pitch system, one independent pitching system per rotor blade with allocated emergency supply

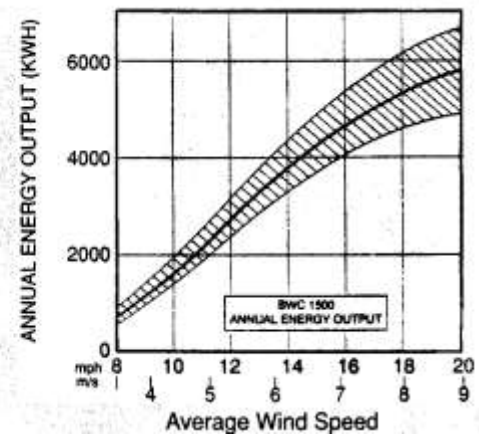
Drive train with generator	
Hub:	Rigid
Main bearings:	Single-row cylindrical roller bearings
Generator:	ENERCON direct-drive synchronous annular generator
Grid feeding:	ENERCON converter
Braking systems:	– 3 independent blade pitch systems with emergency supply – Rotor brake – Rotor lock
Yaw control:	Active via adjustment gears, load-dependent damping
Cut-out wind speed:	28 – 34 m/s (with ENERCON storm control)
Remote monitoring:	ENERCON SCADA



PERFORMANCE AND E

BERGEY BWC 1500

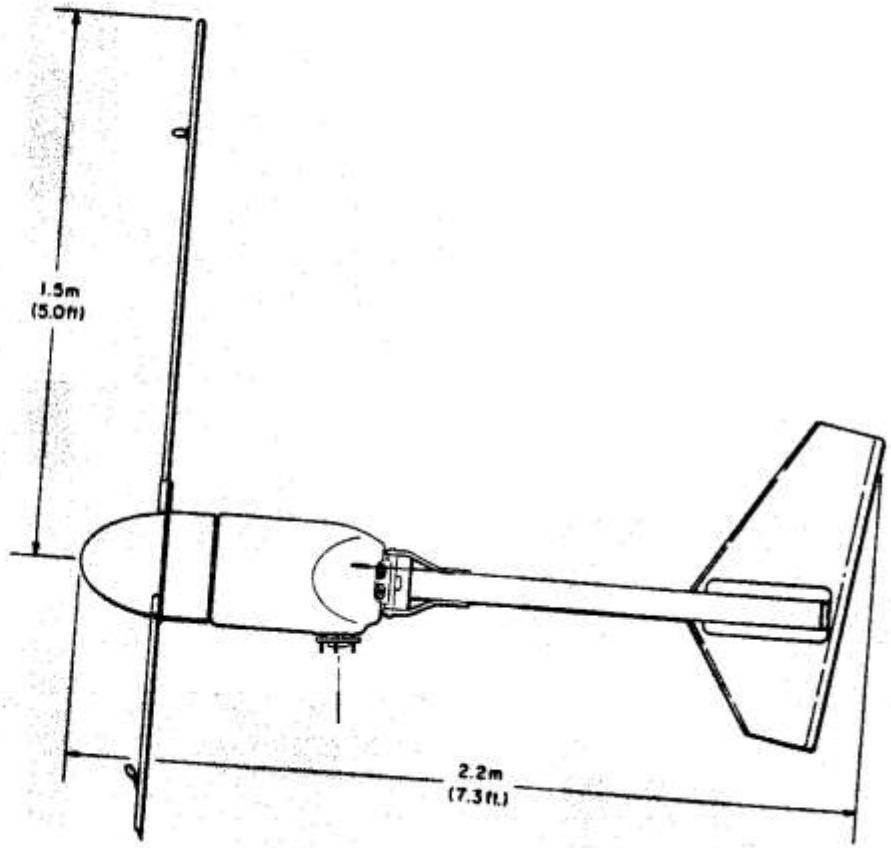




BWC 1500 SPECIFICATIONS

START-UP WIND SPEED. . . . . 8 mph (3.6 m/s)
CUT-IN WIND SPEED. . . . . 8mph (3.6 m/s)
RATED WIND SPEED . . . . . 28 mph (12.5 m/s)
RATED POWER. . . . . 1500 Watts
CUT-OUT WIND SPEED . . . . . None
FURLING WIND SPEED . . . . . 32 mph (14.3 m/s)
MAX DESIGN WIND SPEED. . 120 mph (53.6 m/s)

TYPE . . . . . 3 Blade Upwind
ROTOR DIAMETER . . . . . 10 ft (3.0 m)
WEIGHT . . . . . 168 lbs (76 kg)
BLADE PITCH CONTROL . . . . . POWERFLEX®
OVERSPEED PROTECTION . . . . . AUTOFURL™
DRIVE. . . . . Direct
TEMPERATURE RANGE . . . . . –40° to +140° F
(–40° to +60° C)
GENERATOR . . . . . Permanent Magnet Alternator
OUTPUT FORM . 3 Phase AC, Variable Frequency
WITH VCS-1.5 . . . . . Regulated DC, 12v - 120v



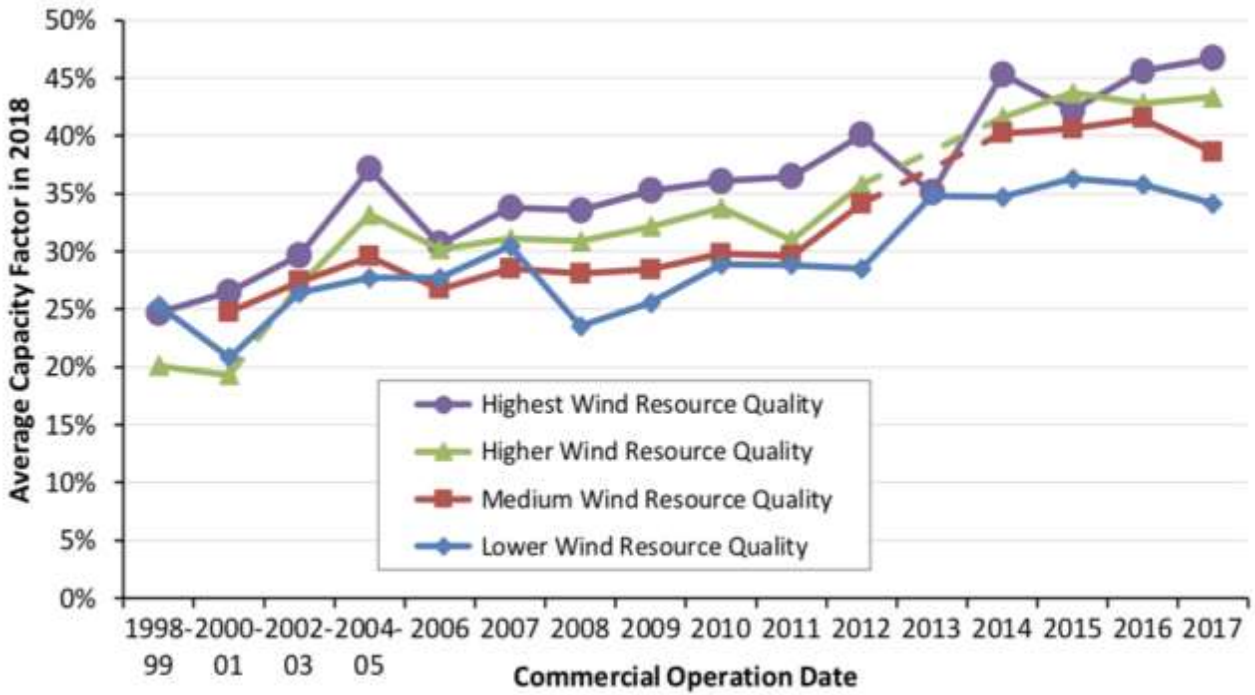
PERFORMANCE AND EFFICIENCY

Capacity Factor:

Capacity factor= Annual yield/rated capacity

Wind turbine can not run all year long (there will times of low or no wind).

U.S. Department of Energy (DOE), 2017 Wind Technologies Market Report



PERFORMANCE AND EFFICIENCY

Capacity Factor=% of full power for all hours of all days
Wind Distribution X Turbine Power Curve=Gross Capacity

Gross Capacity Factor:	**40.0%**
Transformer/Line Losses/Transmission Line	-3.0%
Wake Losses	-4.8%
Control Alogorithm/Turbulence	-1.6%
Blade Contamination	-1.5%
Icing	-1.0%
Turbulence	0.0%
Turbine Availability	-3.0%
Net Capacity Factor [1]	**34.4%**
Gross to Net Conversion	86.0%

Brian Parsons, NREL (http://www1.eere.energy.gov)

WIND POWER CALCULATIONS

What we want?

- To calculate energy from an existing turbine
  - Learn turbine characteristics
  - Learn wind characteristics for that region
  - Calculate energy
- To decide the most suitable turbine for our purpose
  - Determine your energy needs
  - Learn wind characteristics for that region
  - Select turbine by using manufacturers data)
  - Calculate energy
  - Try some others (smaller, larger…) and select the best

WIND POWER CALCULATIONS

There are three method to calculate the output of a wind turbine (or determine the turbine appropriate for the needs and conditions):

- Swept Area method
- Power curve method
- Manufacturer's published estimates for typical wind regimes

WIND POWER CALCU

SWEPT AREA METHOD:

Remember: $P = \frac{1}{2}\rho A v^3$

$(P/A) = P = \frac{1}{2}\rho v^3$ ← Power density

$\rho = 1.225\ kg/m^3$ at 15 °C

$(P/A) = 0.6125\ v^3$

Average (P/A) is available with the wind data (NOTE: Do not calculate from the average speed, they are not the same)

Classes of Wind Power Density at 10 m and 50 m[a]				
	10 m (33 ft)		50 m (164 ft)	
Wind Power Class	Wind Power Density (W/m^2)	Speed[b] m/s (mph)	Wind Power Density (W/m^2)	Speed[b] m/s (mph)
1	<100	<4.4 (9.8)	<200	<5.6 (12.5)
2	100 - 150	4.4 (9.8)/5.1 (11.5)	200 - 300	5.6 (12.5)/6.4 (14.3)
3	150 - 200	5.1 (11.5)/5.6 (12.5)	300 - 400	6.4 (14.3)/7.0 (15.7)
4	200 - 250	5.6 (12.5)/6.0 (13.4)	400 - 500	7.0 (15.7)/7.5 (16.8)
5	250 - 300	6.0 (13.4)/6.4 (14.3)	500 - 600	7.5 (16.8)/8.0 (17.9)
6	300 - 400	6.4 (14.3)/7.0 (15.7)	600 - 800	8.0 (17.9)/8.8 (19.7)
7	>400	>7.0 (15.7)	>800	>8.8 (19.7)

WIND POWER CALCULATIONS

SWEPT AREA METHOD:

Power available in the wind can be found from the power density if the swept area of the turbine rotor is known ($A=\Pi R^2$, R rotor radius)

$$P = (P/A)*A$$

For example:

Rotor radius is 1 m ➔ $A= 3.14\ m^2$

Wind speed 6 m/s ➔ $P/A= 250\ W/m^2$

$$P= (P/A)*A= 785\ W$$

$$E =P*t=785*24*365=785*8760= 6876600\ Wh/year$$

$$=68766\ kWh/year$$

Energy available NOT energy generated!

WIND POWER CALCULATIONS

SWEPT AREA METHOD:

Betz limits: Only 59 % of available energy can be used (theoretically)

Rotor usually deliver 40 %.

After all the losses we receive about 30 % of energy

This number may drop up to 20% in small turbines

Then

Energy generated

$$E=P*t*0.20=68766*0.20 =13753 \text{ kWh/year}$$

WIND POWER CALCULATIONS

POWER CURVE METHOD:

To calculate power of a specific wind turbine for a specific location; we need

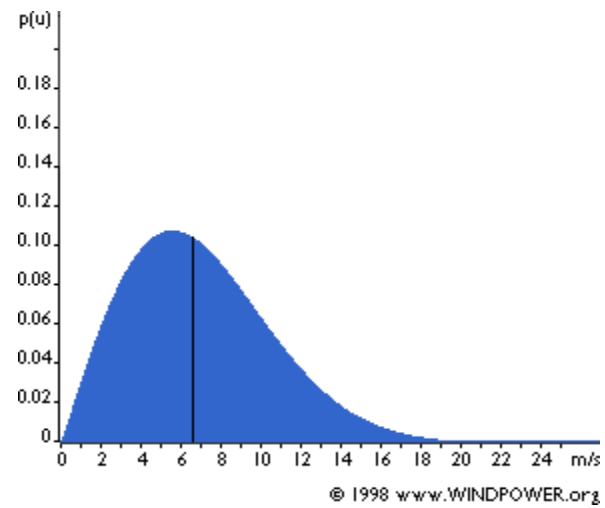
- Manufacturer data sheet for turbine (power curve, tower height etc)
- Wind distribution of location
  - As an histogram, distribution curve or an equation
- Roughness of location

WIND POWER CALCULATIONS

Wind Speed Range (m/s)	Mean Wind Sp (m/s)	Probability (p)
0-1.0	0.5.	0
1.0-2.0	1.5	0.02
.....	.....	.....
	Overall mean =10.7 m/s	Total= 1.00

POWER CURVE METHOD:

1. Make elevation correction for the wind data (using roughness)
2. From cut in speed to cut-out speed, divide the wind speed into small intervals and find the "mean speed" versus "probability" data
   - Get from wind speed distribution table
   - Read from the curve (convert to histogram)
   - Integrate probability distribution function in a small interval



$$f(x;\lambda,k) = \begin{cases} \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

WIND POWER CALCULATIONS

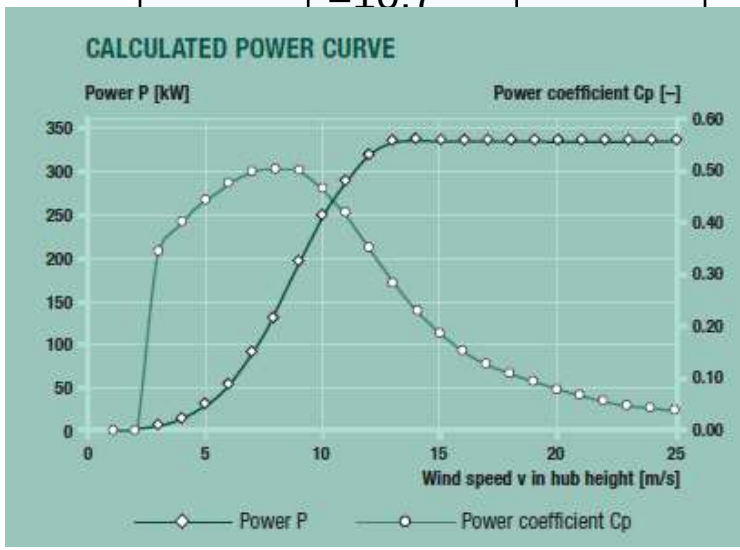
POWER CURVE METHOD:

3. Find power for each wind interval (use mean) from power curve of turbine (P_1, P_2 , P_2....)
4. Multiply by the probability ($P_1 x p_1$,....)
5. Add all to find mean power generated

   $P_{mean}=P_1 x p_1+P_2 x p_2+$.....
6. Multiple by hours in a year to get total energy produced
7. Consider losses to get actual energy
8. Repeat procedure for smaller &larger turbine since they may suit better

Note: $P_{mean} \neq P$ at mean speed because $(v^3)_{mean} \neq (v_{mean})^3$

Wind Speed Range (m/s)	Mean Wind Sp (m/s)	Probability (p)
0-1.0	0.5.	0
1.0-2.0	1.5	0.02
.....	.....	.....
	Overall mean =10.7	Total= 1.00



WIND POWE

POWER CURVE METHOD: EXAMPLE

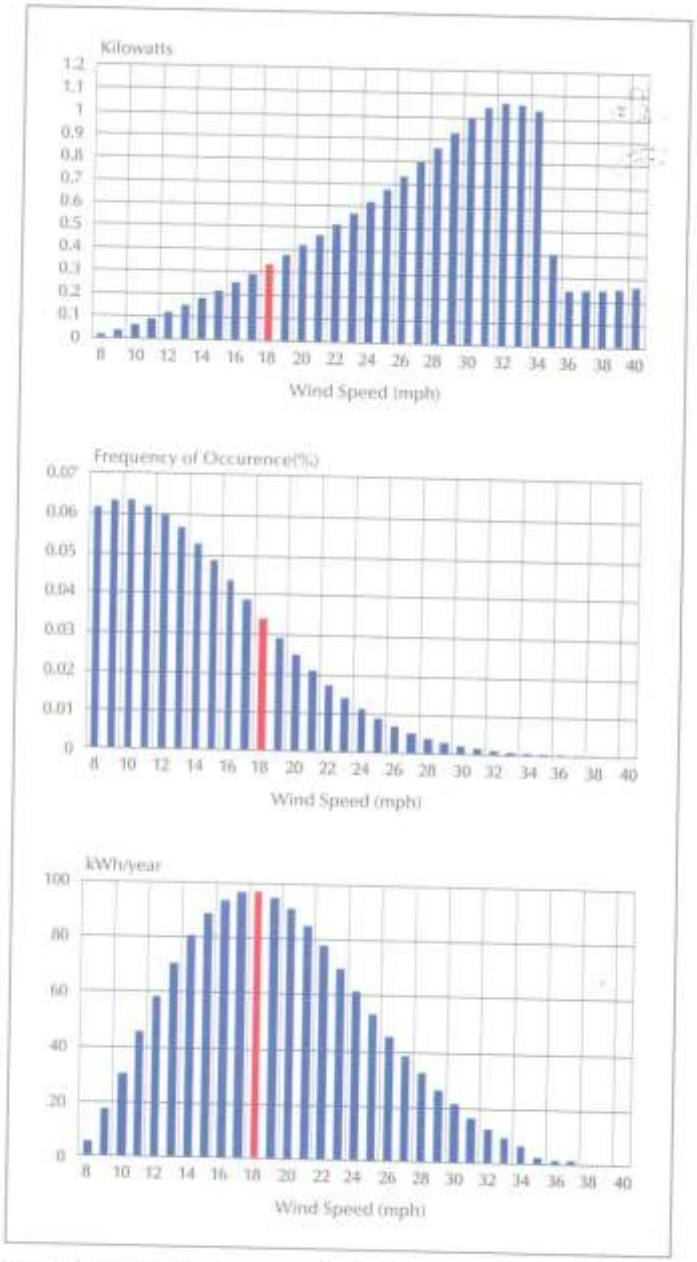


Figure 4-6. Power curve method of calculating annual energy output. Power is shown on the vertical axis, and wind speed at hub height on the horizontal axis. In this example the wind turbine produces slightly more than 300 watts at a wind speed of 18 mph (8 m/s). Winds of this speed occur 0.0335 percent of the time or about 300 hours per year. Consequently, this wind speed bin contributes almost 100 kWh per year of the turbines' total annual generation.

WIND P

POWER CURVE METHOD: EXAMPLE

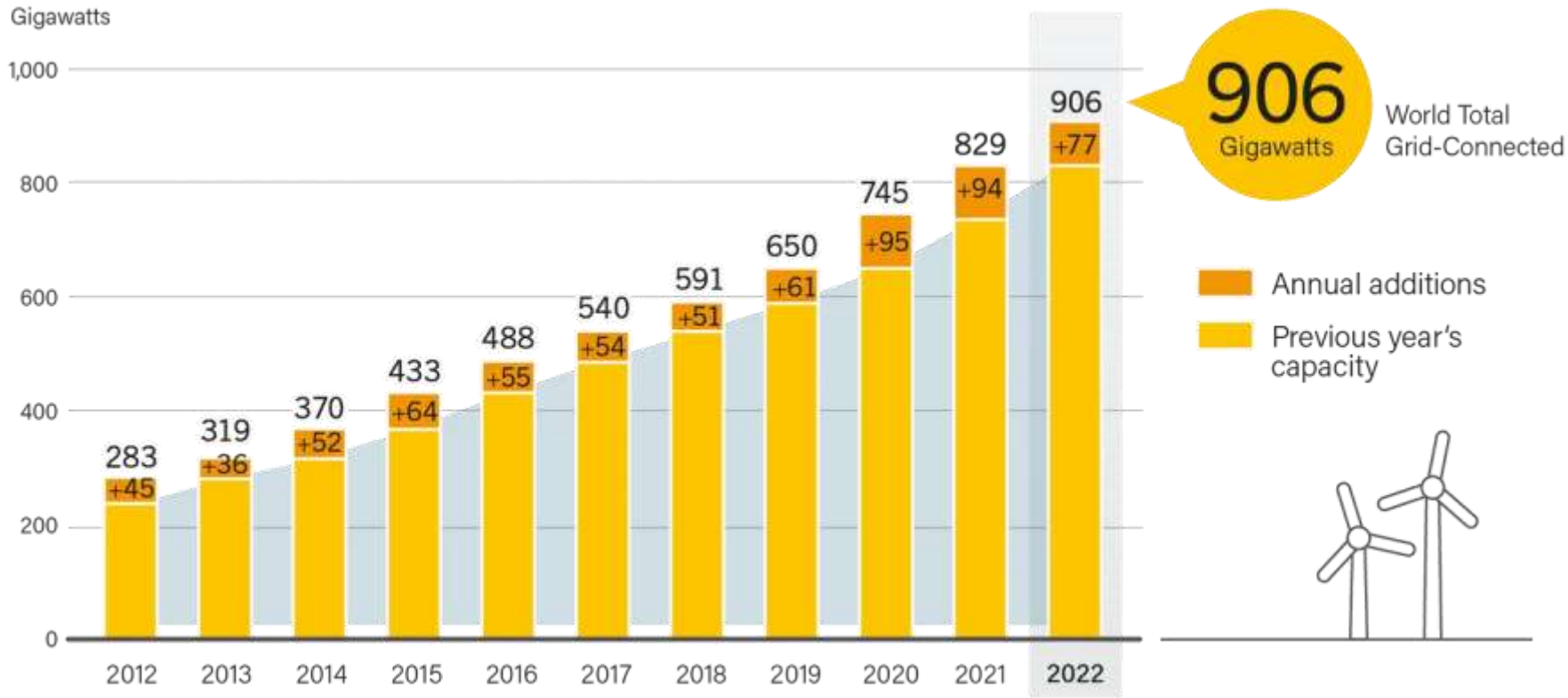
Table 4-2
Estimating Annual Energy Output Using a Power Curve

Wind Speed Bin (mph)	Instantaneous Power (kW)	Rayleigh Frequency Distribution (% Occurrence)	(h/yr)	Energy (kWh/yr)
8	0.010	0.0616	539	5
9	0.030	0.0631	553	17
10	0.055	0.0632	554	30
11	0.082	0.0620	543	45
12	0.111	0.0597	523	58
13	0.142	0.0564	494	70
14	0.175	0.0524	459	80
15	0.210	0.0480	420	88
16	0.247	0.0432	378	93
17	0.286	0.0383	336	96
18	0.327	0.0335	294	96
19	0.370	0.0289	253	94
20	0.415	0.0246	216	90
33	1.055	0.0009	8	9
34	1.030	0.0007	6	6
35	0.400	0.0005	4	2
36	0.240	0.0003	3	1
37	0.250	0.0002	2	1
38	0.240	0.0002	1	0
39	0.250	0.0001	1	0
40	0.260	0.0001	1	0
Annual Energy Output (AEO)				1,416

Note: Sample small wind turbine 2.4 meters (8 ft) in diameter, Rayleigh distribution average annual wind speed = 12 mph

FACTS AND TRENDS

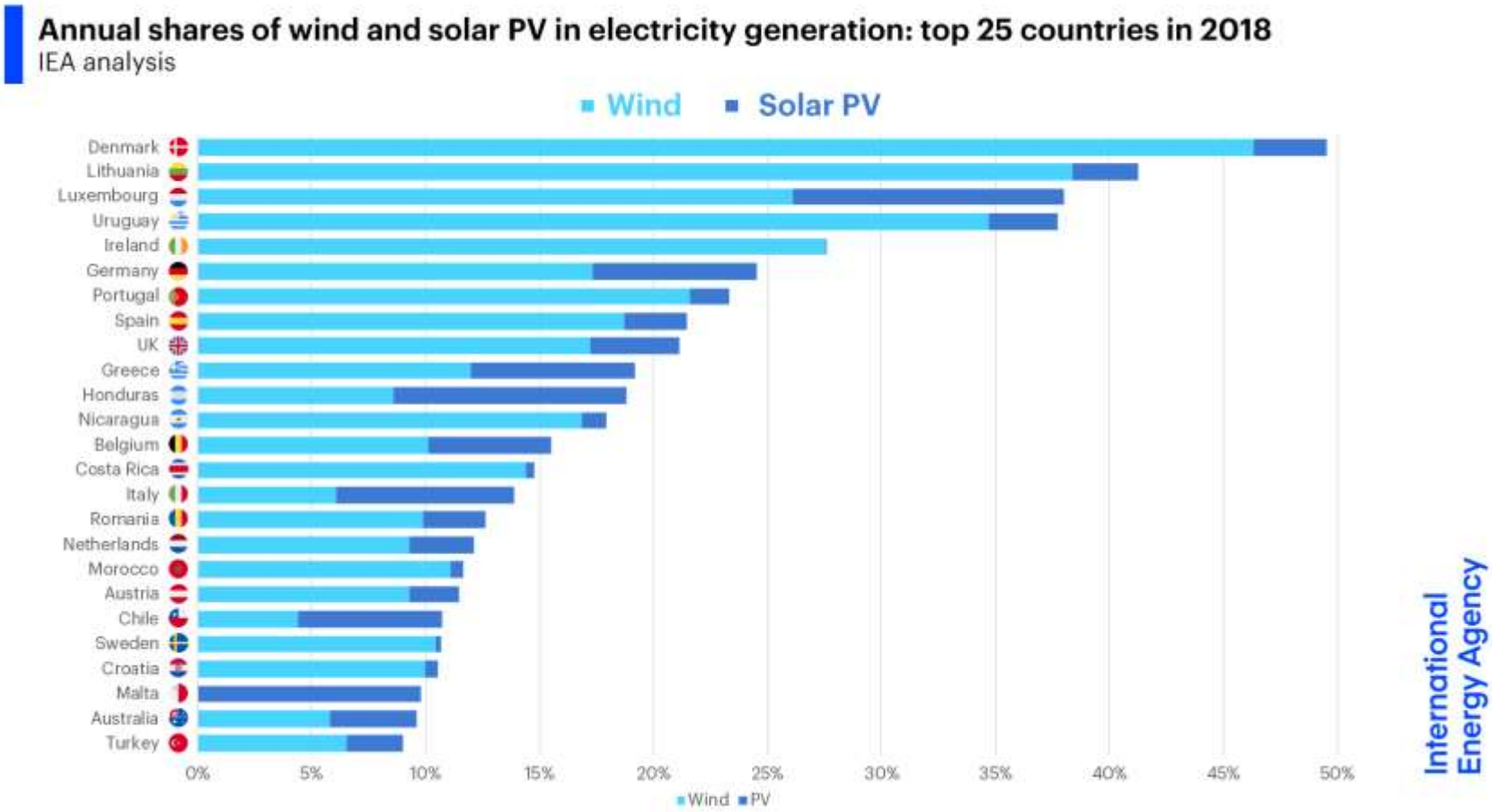
REN21 RENEWABLES 2023 GLOBAL STATUS REPORT



FACTS AND TRENDS

- An estimated 89 GW of wind power capacity was mechanically installed in 2022, of which more than 77 GW was added to the world's grids, bringing the total grid-connected capacity to an estimated 906 GW.
- Global grid-connected additions fell more than 17% due mainly to slowdowns in China and the United States; Europe was the only region where installations rose.
- Countries around the world increased their wind power targets, driven by climate change, energy security, and economic growth goals, as well as the cost-competitiveness of wind energy.
- While offshore installations declined relative to 2021, due mainly to a temporary slowdown in China, the global pipeline nearly doubled in 2022 to nearly 1.2 terawatts across 38 countries.
- The industry continued to innovate to change the cost base of projects; to address challenges associated with scaling up production, transport and other logistical issues; and to enhance the value of wind energy while further improving its environmental and social sustainability.

FACTS AND TRENDS



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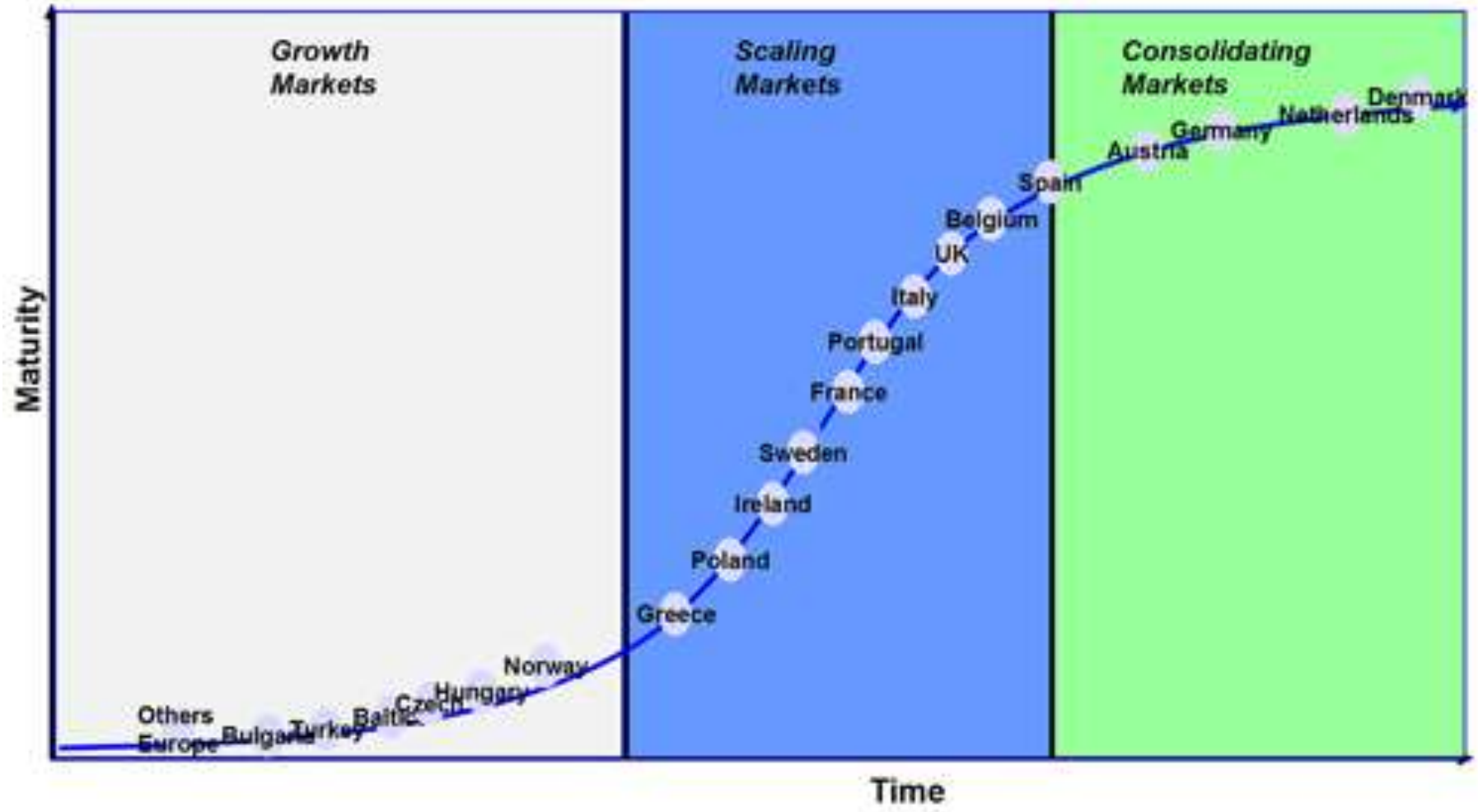
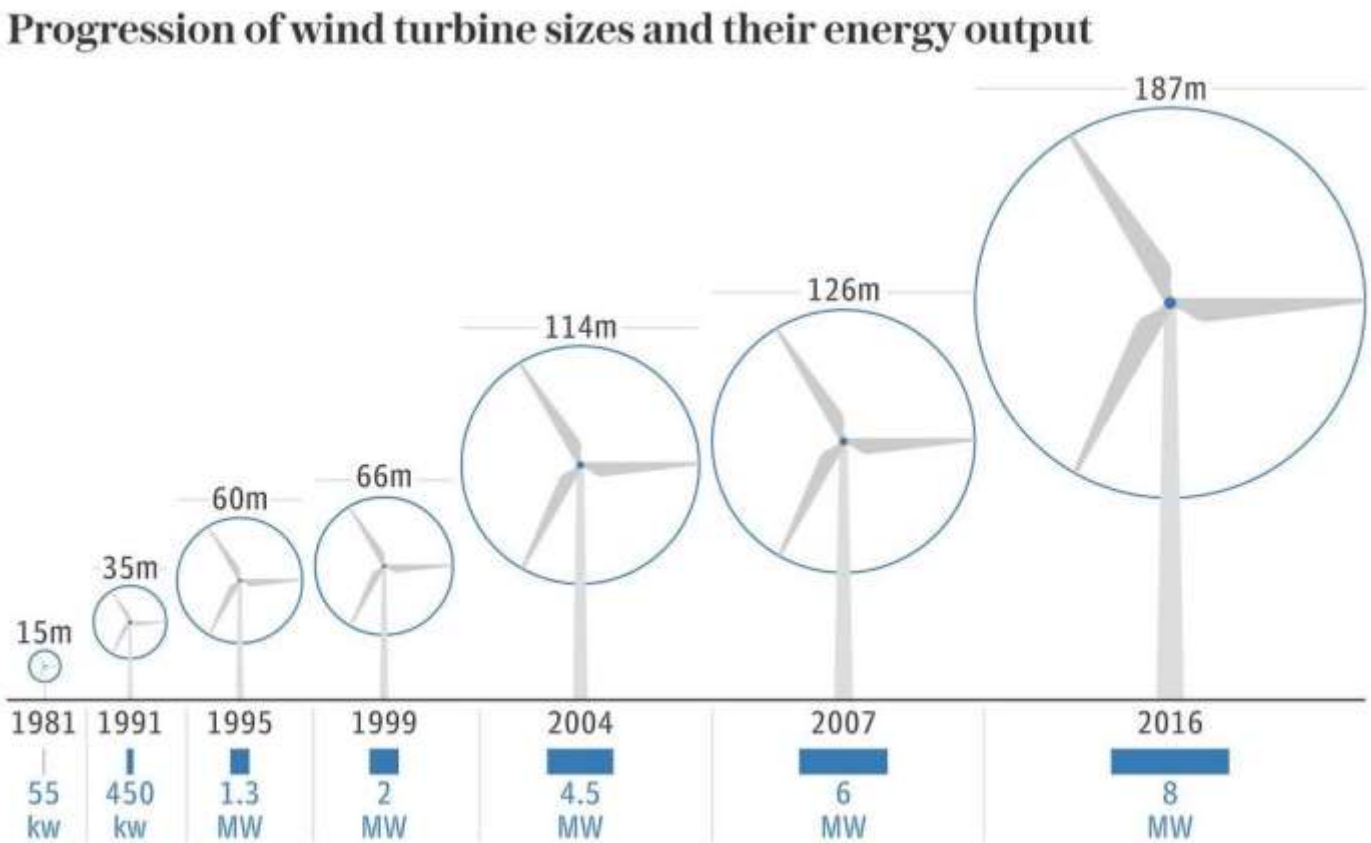


The Wind Energy Facts,EWEA

FACTS AND TRENDS



Published in Proceedings of the IEEE 2015

High-power wind energy conversion systems: State-of-the-art and emerging technologiesV. YaramasuBin WuP. SenS. KouroM. Narimani

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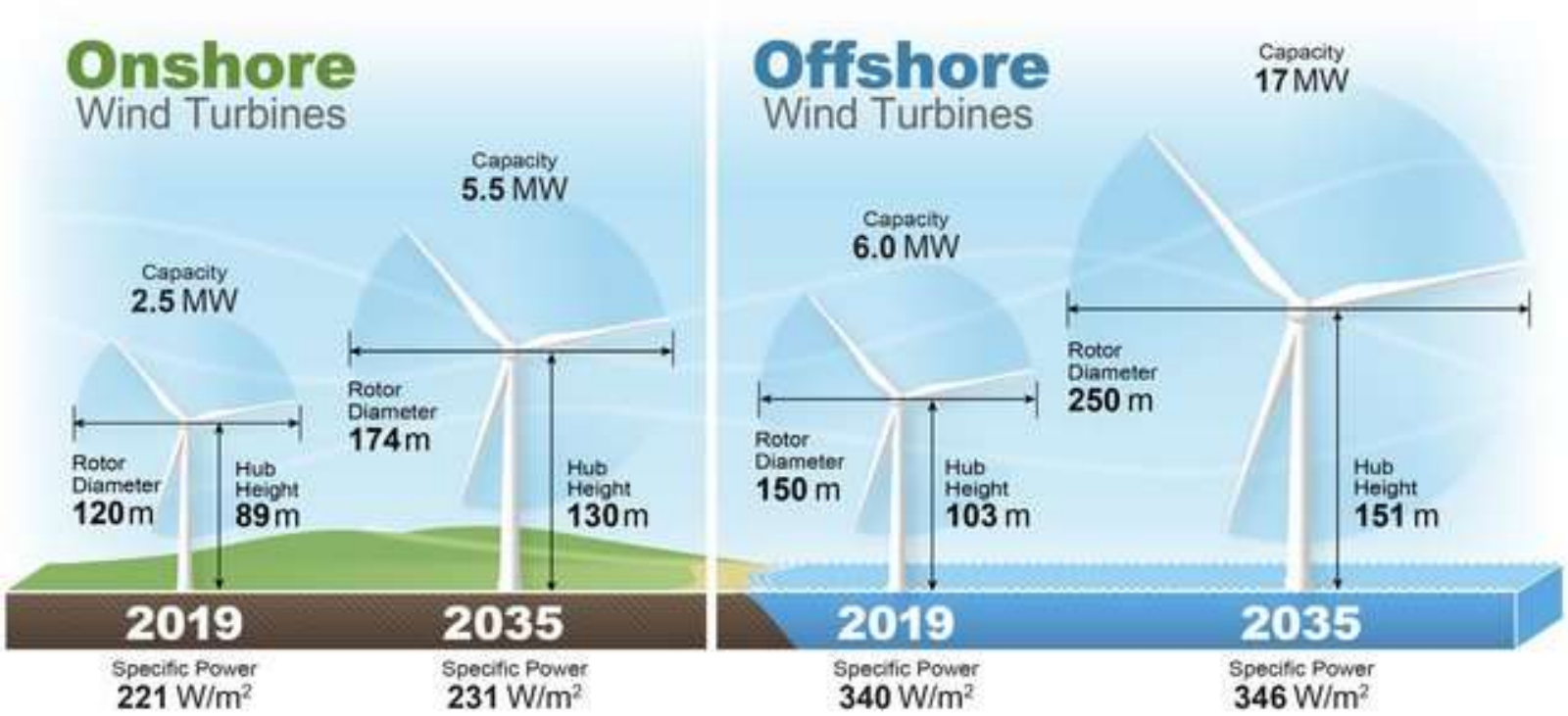

DOI: 10.1038/s41560-021-00810-z

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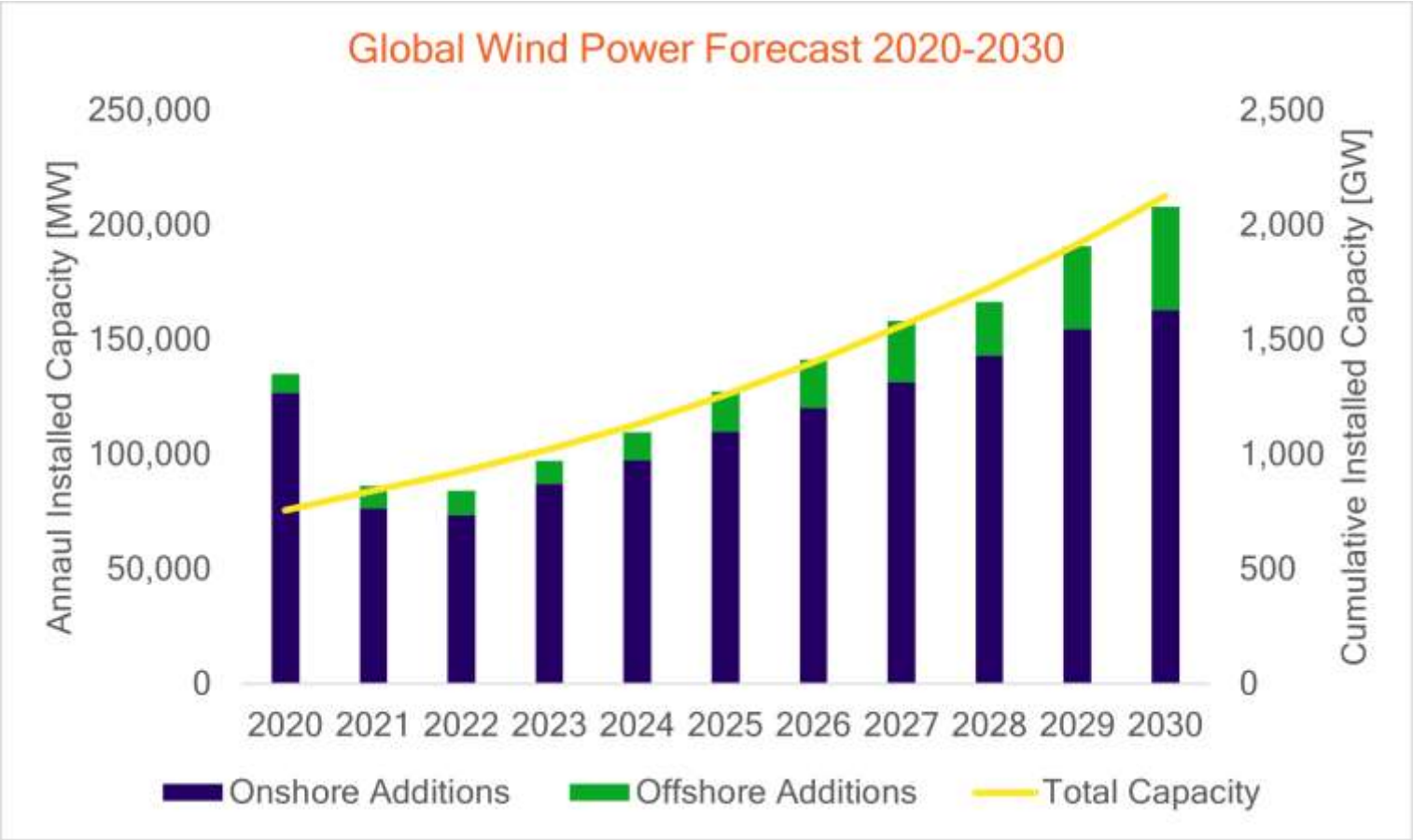


https://www.enlit.world/renewable-energy/wind/wind-to-account-for-two-thirds-of-global-power-production-by-2030/

FACTS AND TRENDS

https://doi.org/10.1016/j.seta.2022.102766

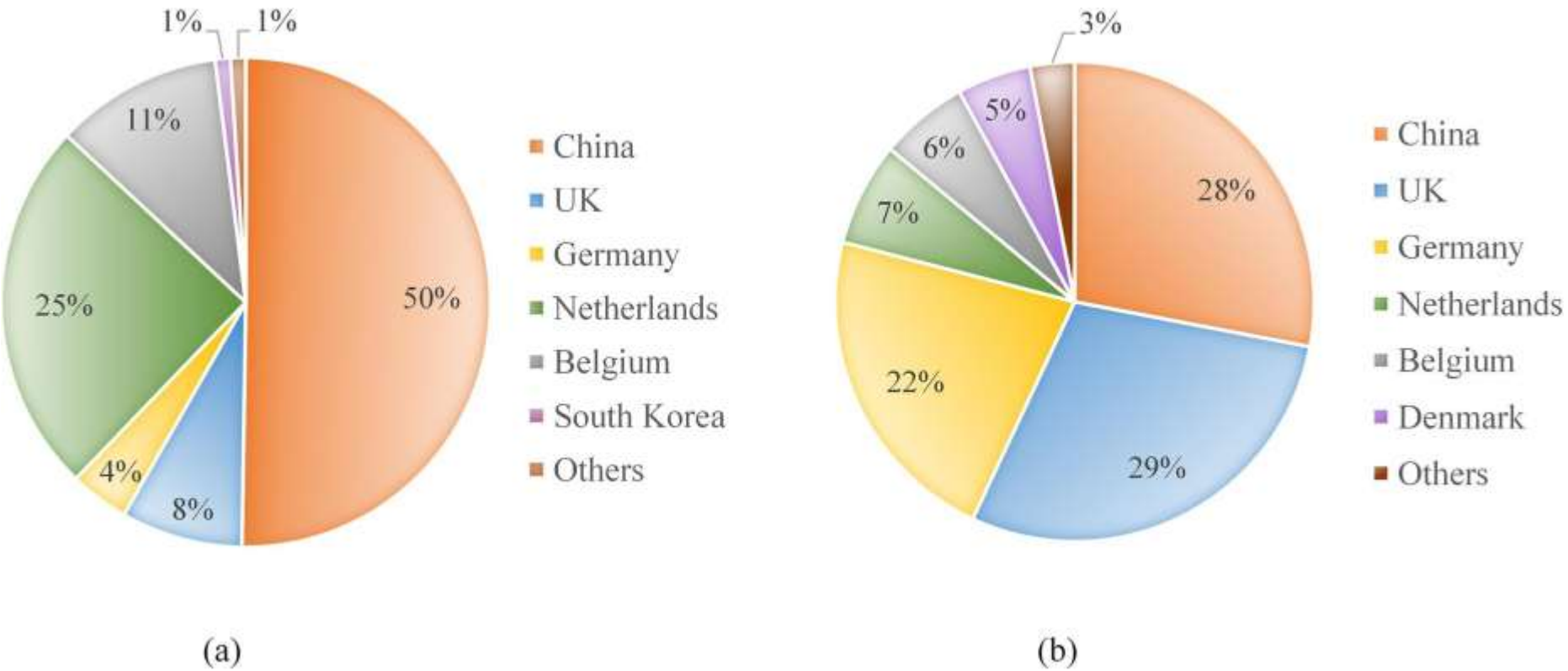


Fig. 2. Global installed offshore wind power capacity in 2021 *Source: Global Wind Energy Council (GWEC)*. (a) New installations. (b) Total installations.

FACTS AND TRENDS

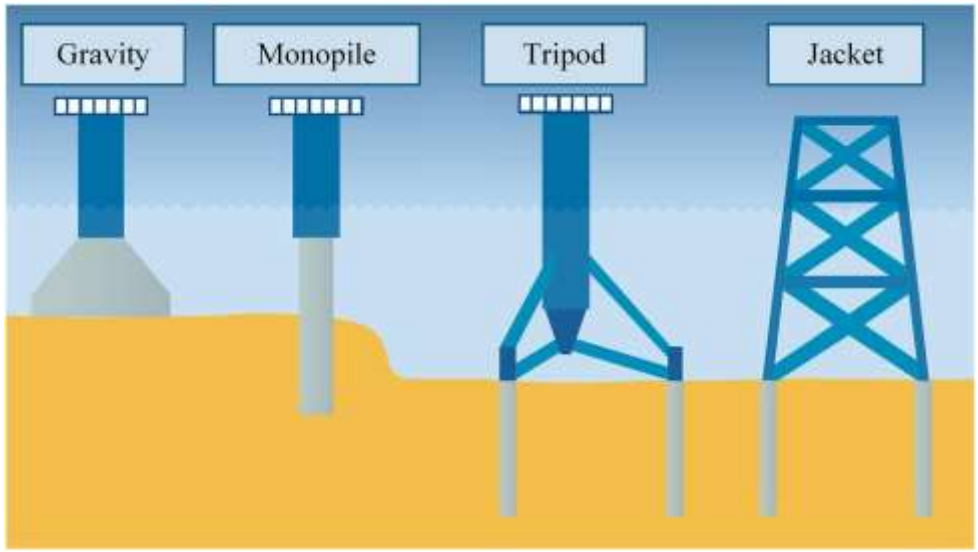
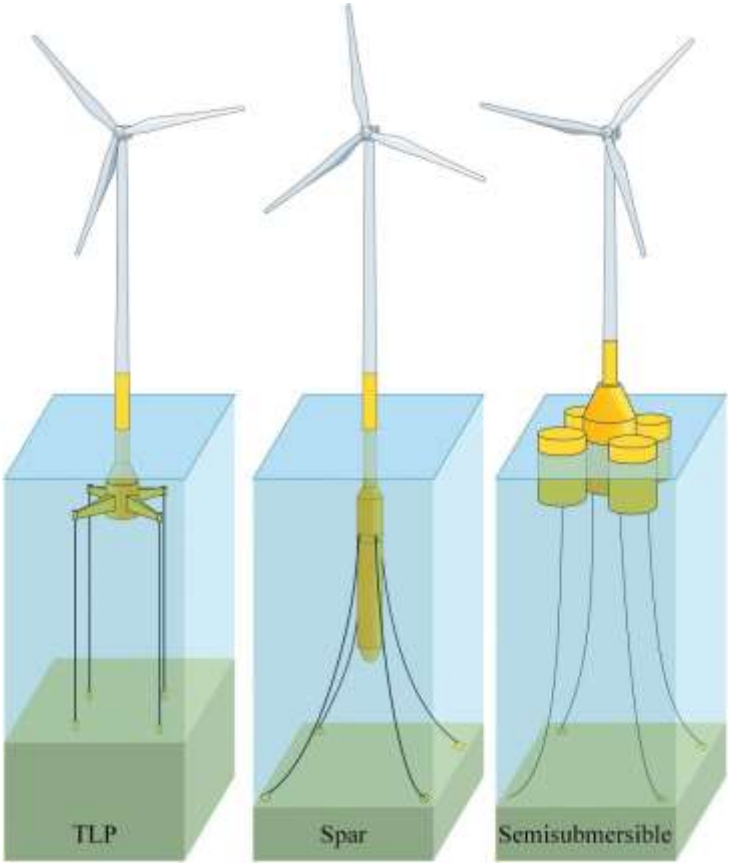


Fig. 7. Fixed offshore wind foundation structures *Source: International Renewable Energy Agency (IRENA)*.

https://doi.org/10.1016/j.seta.2022.102766

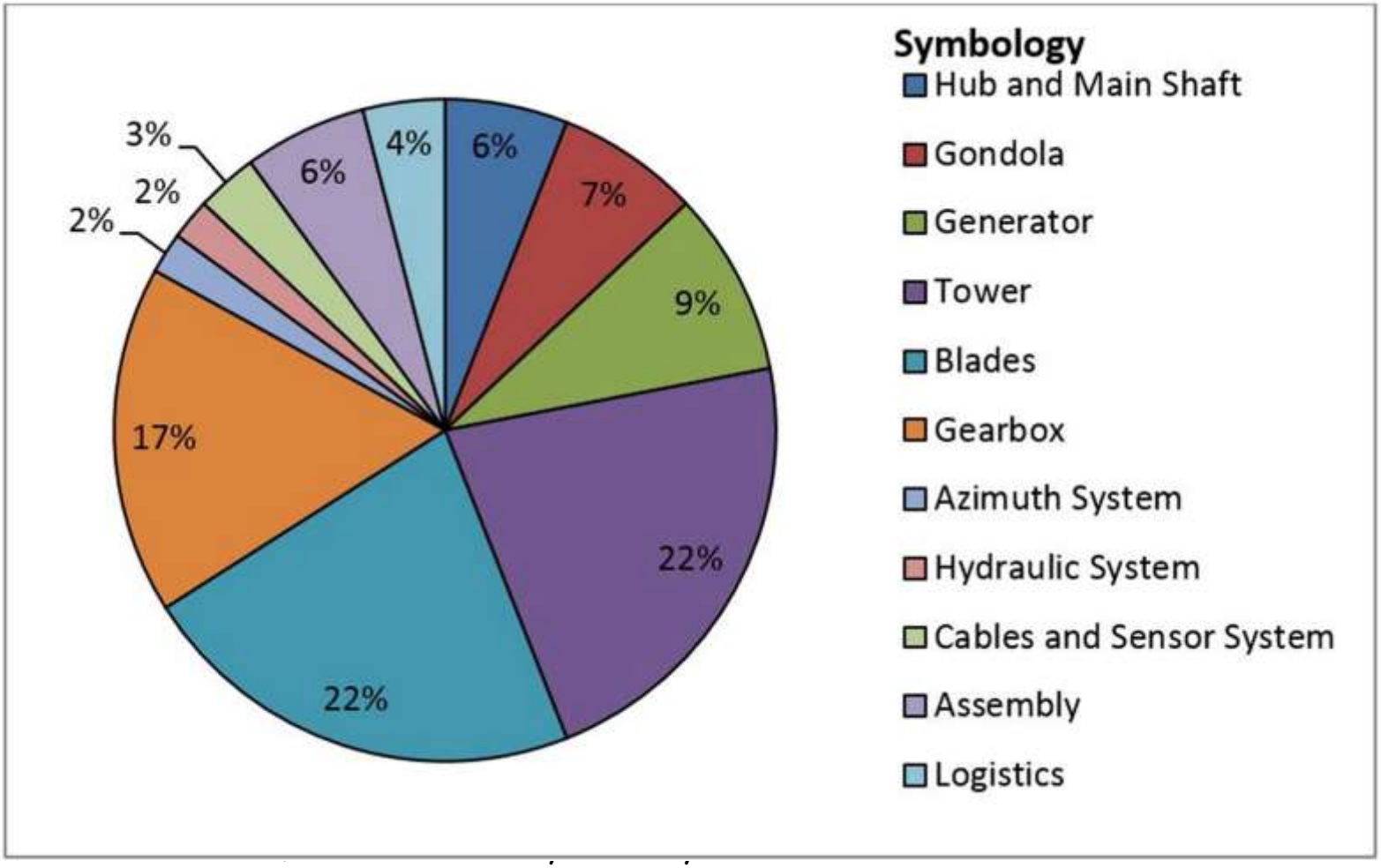
Fig. 8. Floating offshore wind foundation structures *Source: Aquatic Renewable Energy Technologies (Aqua-RET)*.



ECONOMICS

Turbine Cost:

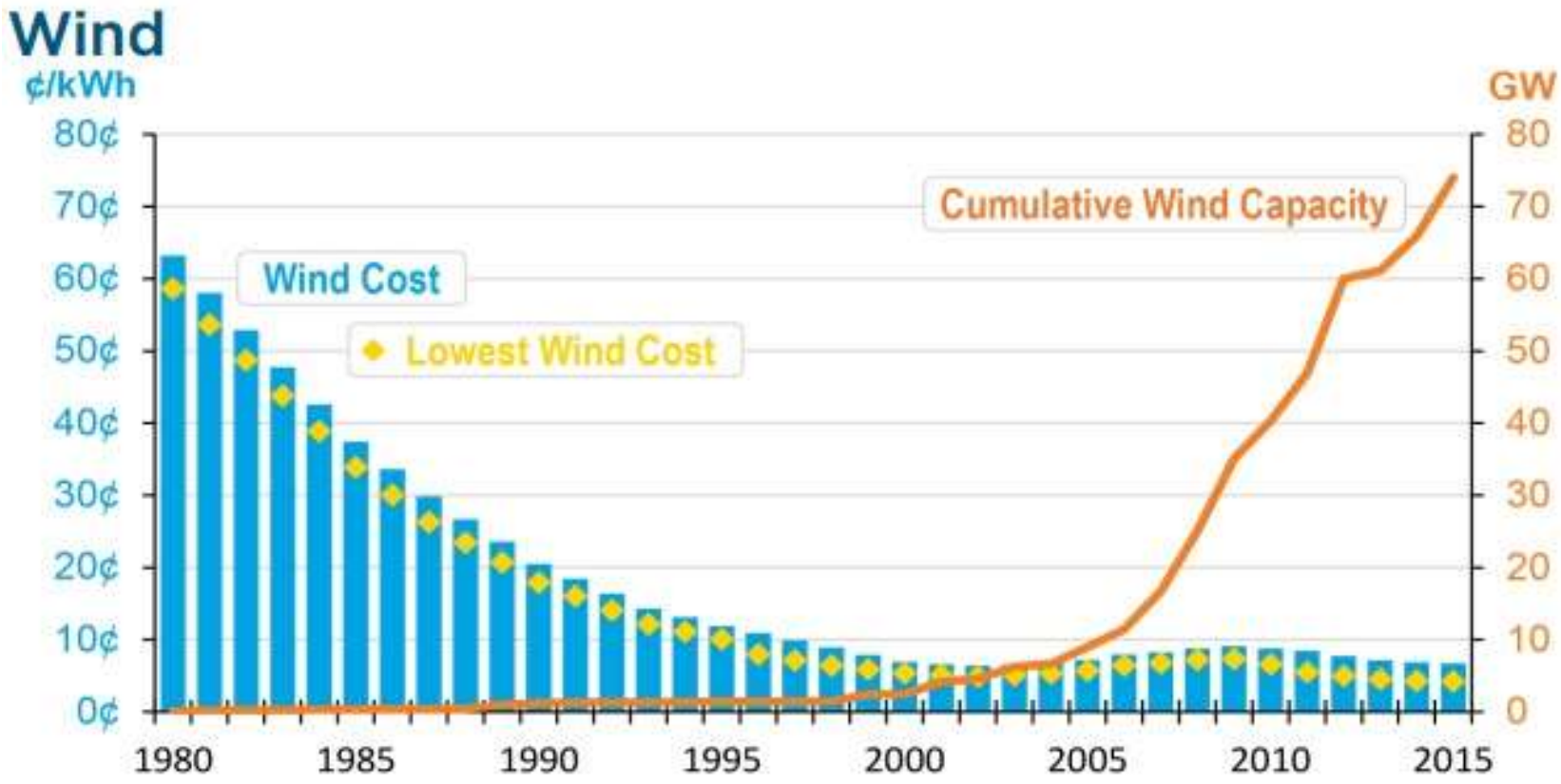
https://doi.org/10.1016/j.enpol.2019.05.046



ECONOMICS

Electricity Cost (USA)

https://www.energy.gov/eere/articles/revolutionnow-rewind-illustrating-wind-energy-story



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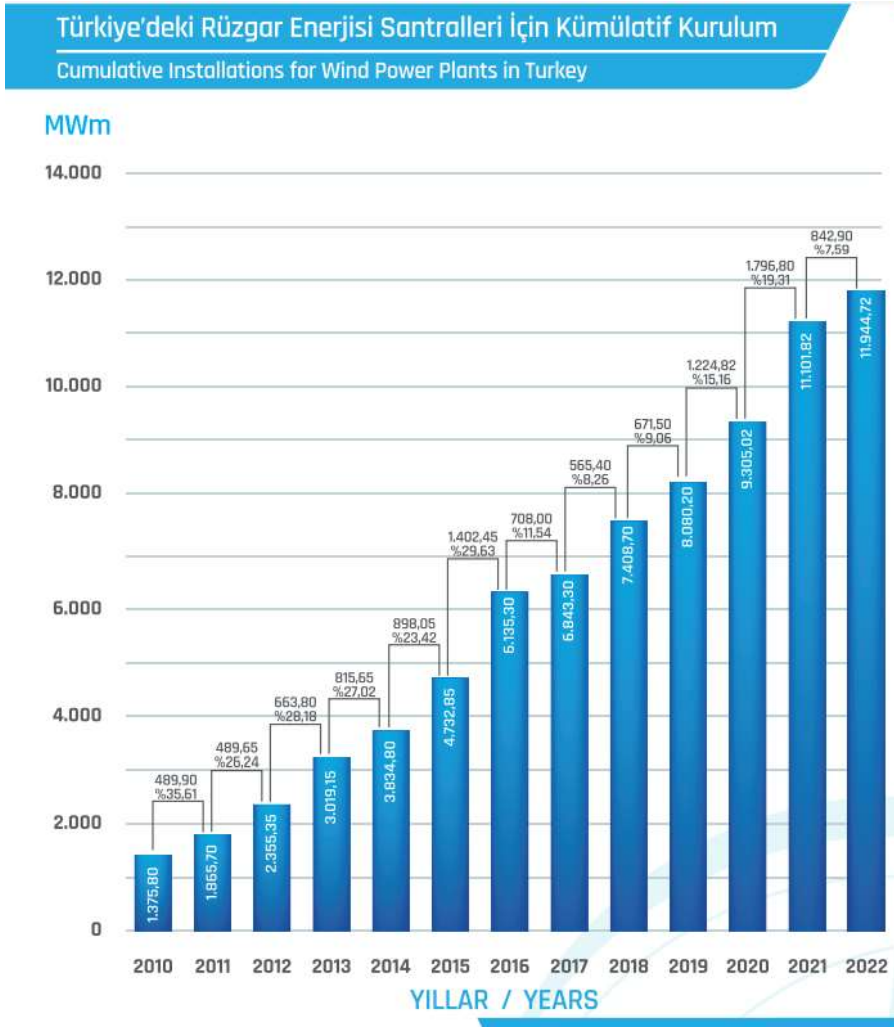

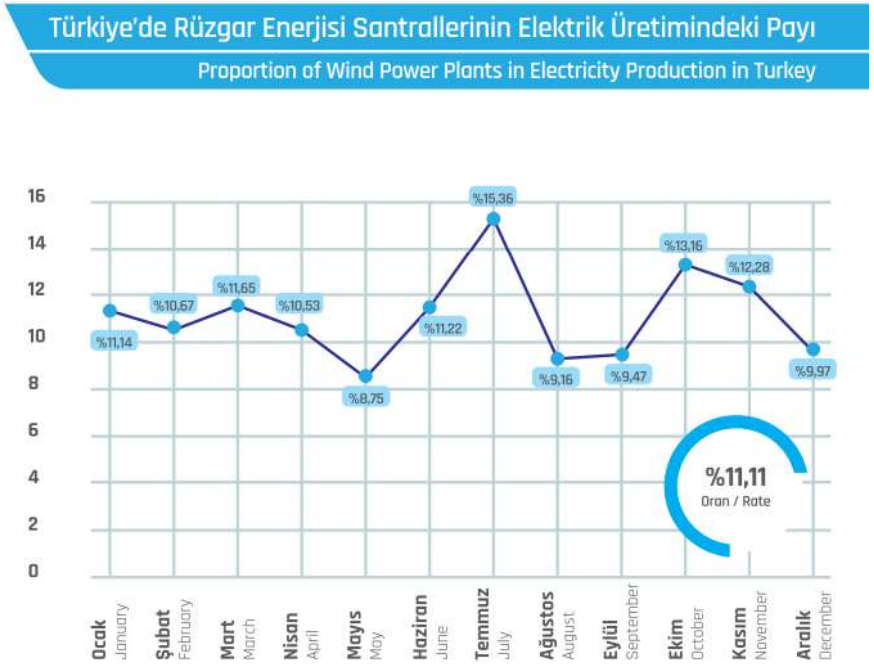

Source: TUREB

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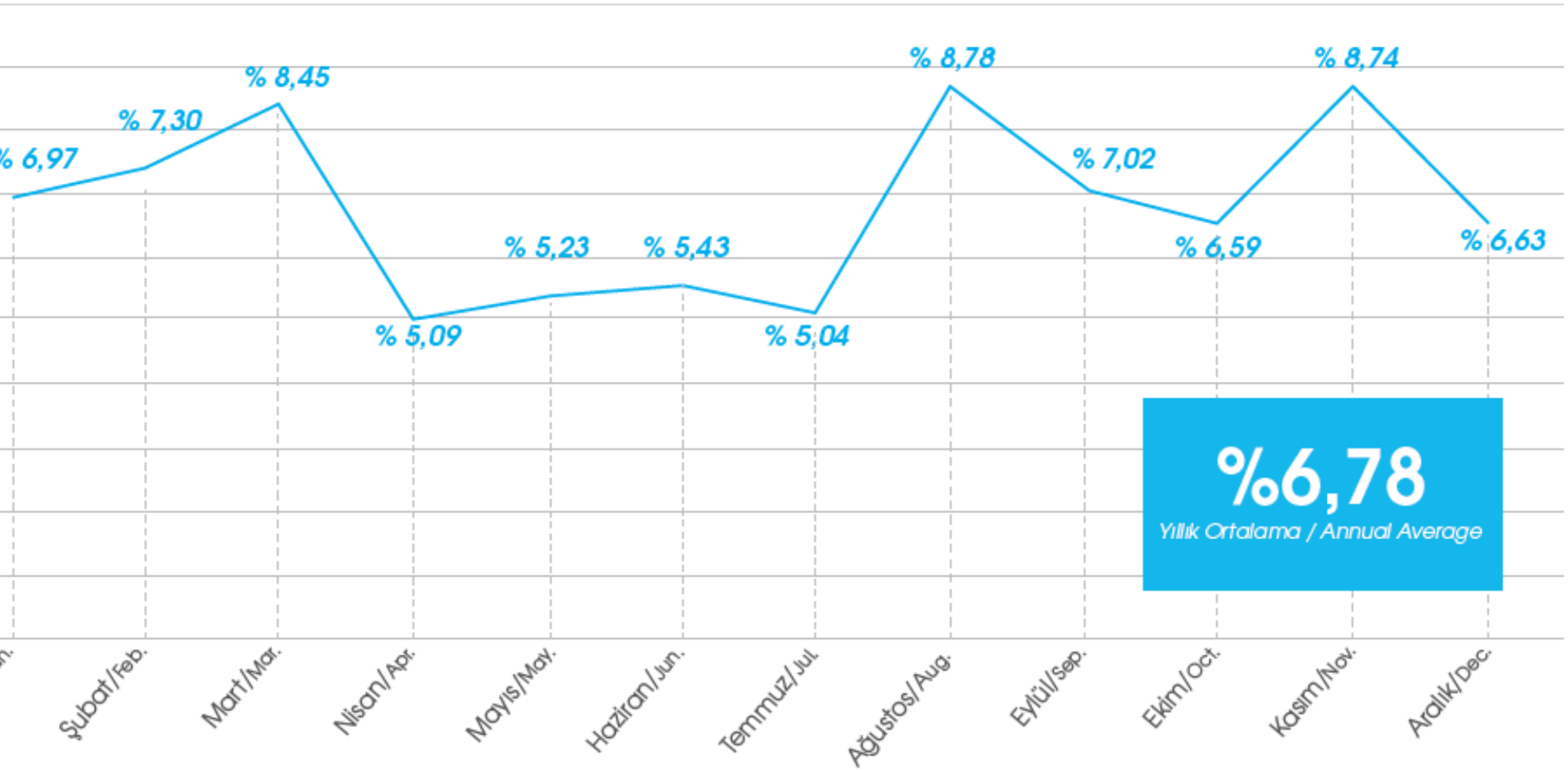
he share of WPP's in Electricity Generation in Turkey



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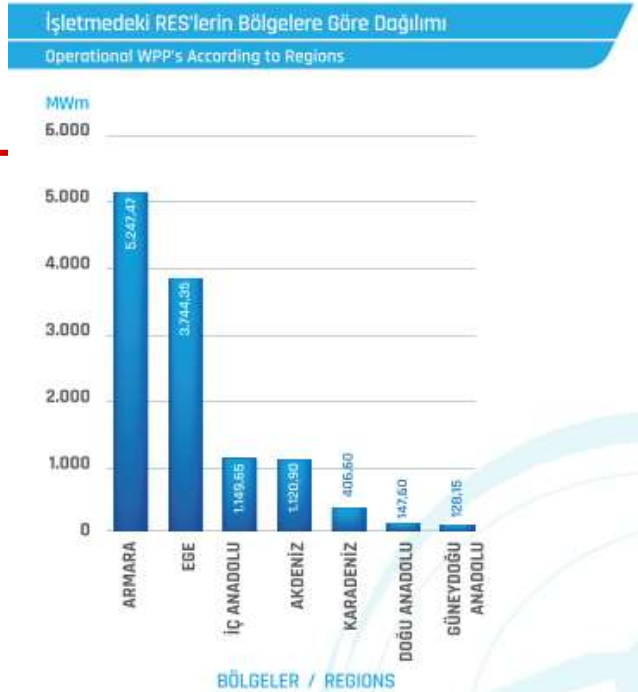

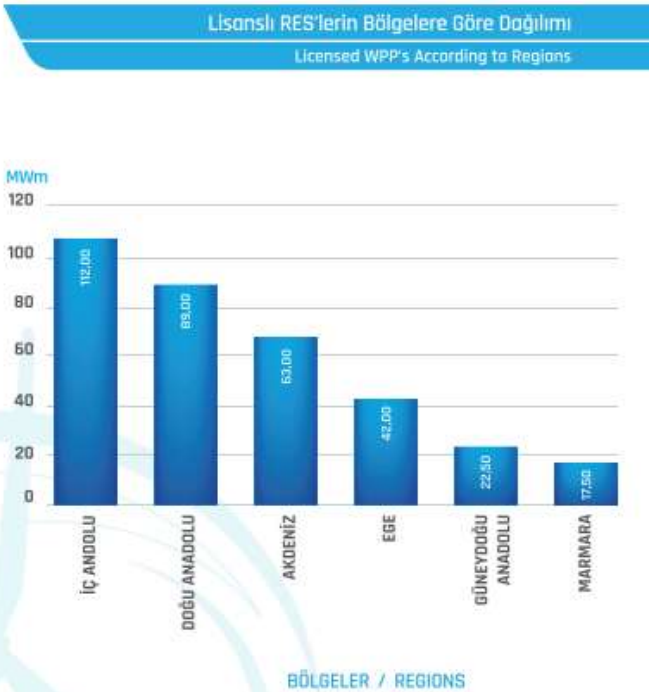

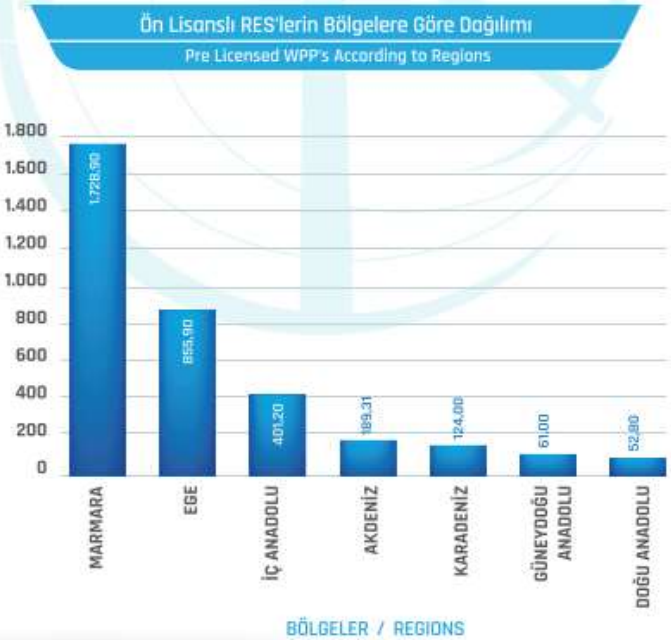

Source: TUREB

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